A Project report on

Multi-Stage Machine Learning and Deep Learning Pipeline for Precise Oil Spill Detection and Classification

A Dissertation submitted to JNTUH, Hyderabad in partial fulfillment of the academic requirements for the award of the degree.

Bachelor of Technology

in

Computer Science and Engineering

(Artificial Intelligence & Machine Learning)

Submitted by

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CERTIFICATE

This is to certify that the Major Project report entitled **" Multi-Stage Machine Learning and Deep Learning Pipeline for Precise Oil Spill Detection and Classification"**being submitted by GUNDLAPALLI RAJ KUMAR (22H51A6626), KANTAREDDY SATHVIK REDDY (22H51A6689) in partial fulfillment for the award of **Bachelor of Technology in Computer Science and Engineering (AI&ML)** is a record of bonafide work carried out under my guidance and supervision.

The results embodies in this project report have not been submitted to any other University or Institute for the award of any Degree.

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DECLARATION

We hereby declare that results embodied in this Report of Project on "Multi-Stage Machine Learning and Deep Learning Pipeline for Precise Oil Spill Detection and Classification" are from work carried out by using partial fulfillment of the requirements for the award of Bachelor of Technology in **Computer Science and Engineering (AI&ML)**. We have not submitted this report to any other university/institute for the award of any other degree.

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ABSTRACT

Oil spills are among the most serious threats to marine and coastal ecosystems. Early and accurate detection is essential for timely response and minimizing environmental damage. This project proposes a hybrid oil spill detection and classification system using Synthetic Aperture Radar (SAR) imagery. The method begins with SAR image denoising using the Lee Filter, followed by Adaptive Histogram Thresholding for segmentation, and classification using a Custom Convolutional Neural Network (CNN) with Softmax Activation. The solution integrates a web-based interface using Streamlit/Flask for usability. By combining strengths from multiple existing techniques, this system enhances accuracy, efficiency, and scalability, making it ideal for real-time environmental monitoring and practical student-level implementation.

CHAPTER 1
INTRODUCTION

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INTRODUCTION

1.1 Problem Statement

There is a need for a lightweight, accurate, and modular system that can preprocess, segment, and classify oil spill areas from SAR images in real time while being deployable through a simple interface. This project addresses these issues using a combination of established techniques, optimized for efficiency and practical application. Additionally, the system is designed to handle large volumes of satellite data with minimal computational overhead, ensuring faster processing and timely detection. It also focuses on scalability, allowing integration with existing environmental monitoring platforms and decision-support systems. Furthermore, the approach emphasizes robustness against noise and varying imaging conditions, improving reliability in real-world maritime surveillance scenarios.

1.2 Research Objective

- **Objective-1**: Data Collection and Preprocessing:
  SAR image datasets were gathered, and image normalization, denoising, or and enhancement were applied to provide clean model input.
- **Objective-2**: Model Development and Combination:
  To increase detection accuracy, two CNN-based models were constructed and integrated into a single improved architecture.
- **Objective-3**:Training, Tuning, and Evaluation:
  The model was trained with optimal hyperparameters and assessed using F1-score, precision, and recall metrics.
- **Objective-4**: Deployment and Validation:
  The finished model was deployed via an intuitive web interface for real-time use after being validated on unseen SAR images.

1.3 Project Scope and Limitations

1.3.1 Project Scope

This project focuses on developing an efficient system for processing and analyzing Synthetic Aperture Radar (SAR) imagery, specifically targeting detection tasks such as object or anomaly identification in remote sensing applications. The scope is limited to SAR data handling, excluding other imaging modalities like optical or multispectral data to ensure a lightweight and modular architecture. Key activities include acquiring and preprocessing publicly available SAR datasets, designing and integrating CNN-based models with data augmentation techniques, training and evaluating the system against established benchmarks, deploying it for real-time use via a web interface, and validating its performance on unseen data. The project emphasizes computational efficiency, robustness to noise inherent in SAR images, and practical applicability in scenarios like environmental monitoring or disaster response, while avoiding complex multi-sensor fusion or hardware-specific optimizations.

1.3.2 Limitations

- **Data Limitation**: Relies solely on publicly available SAR datasets; excludes real-time or proprietary satellite data.
- **Model Limitation**: Uses only CNN-based architectures; no transformers or large-scale models for lightweight deployment.
- **Hardware Limitation**: Optimized for standard CPU/GPU systems; not suited for edge devices or embedded platforms.
- **Deployment Limitation**: Web interface via Streamlit/Flask only; no mobile, API, or batch processing support.
- **Generalization Limitation**: Tested on similar SAR sensor conditions; limited cross-sensor or seasonal robustness.

CHAPTER 2
LITERATURE SURVEY

CHAPTER 2
LITERATURE SURVEY

2.1 Machine Learning for the Detection of Oil Spills (Kubat, Holte, & Matwin)

2.1.1 Introduction

The study *"Machine Learning for the Detection of Oil Spills"* by Kubat, Holte, and Matwin represents one of the earliest applications of classical machine learning techniques to environmental monitoring. The primary objective was to develop a system capable of automatically identifying oil spills in radar satellite images, distinguishing them from lookalike phenomena such as low-wind areas, algal blooms, or calm sea regions. Each radar image is divided into multiple segments, and every segment is described by a set of numerical features including brightness, area, shape descriptors, and texture characteristics. These feature vectors serve as inputs to machine learning models trained to classify each segment as either **"Oil Spill"** or **"Non-Oil"**.

- The researchers employed several algorithms for this binary classification task:
- **C4.5 Decision Tree Algorithm:** to generate interpretable classification rules based on feature thresholds.
- **1-Nearest Neighbor (1-NN):** to classify regions based on similarity to known examples.
- **SHRINK Algorithm:** a custom-designed model aimed at handling class imbalance, where oil spill samples are significantly fewer than non-oil samples.
- All experiments were conducted using the **Canadian Environmental Hazards Detection System (CEHDS)**, a specialized platform for environmental image analysis. As this research predates modern deep learning frameworks, the implementation relied entirely on classical ML principles and in-house algorithmic development.

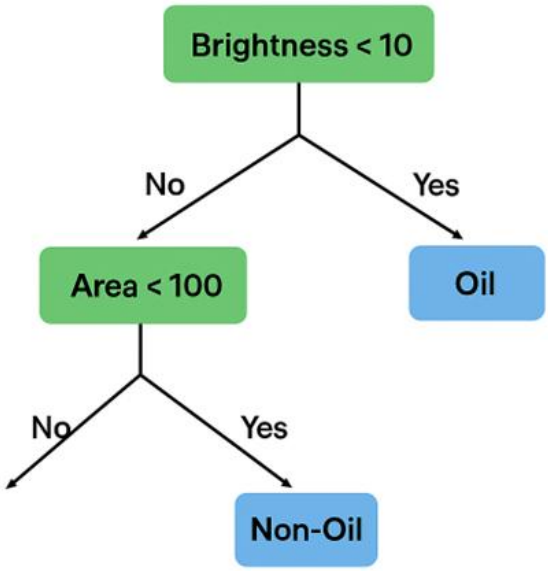

Figure 2.1.1 Decision Tree

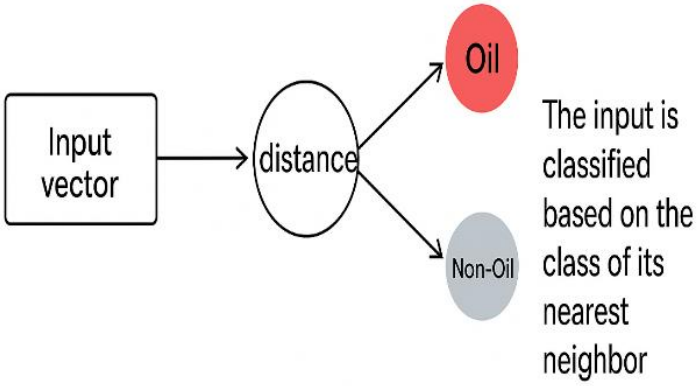

Figure 2.1.2 Nearest Neighbour Algorithm

2.1.2 Merits, Demerits and Challenges

Merits

- Demonstrated one of the first successful uses of machine learning in environmental remote sensing.
- Provided interpretable models (especially through decision trees) that helped experts understand classification criteria.
- Effective on small datasets, since classical algorithms like C4.5 and 1-NN do not require massive data for training.
- Introduced SHRINK, an early approach to address imbalanced data problems in environmental monitoring.

Demerits

- Performance is limited by hand-crafted features, which may not fully capture the complex patterns of radar imagery.
- 1-NN suffers from high computational cost during prediction, as it must compare against all training examples.
- Decision trees can overfit noisy or small datasets if not properly pruned.
- Lack of automated feature extraction compared to modern deep learning methods

Challenges

- Class imbalance — oil spills are rare compared to lookalikes, making learning difficult.
- Feature selection and engineering were crucial but manual, requiring significant domain expertise.
- Limited computational resources and absence of modern frameworks constrained experimentation.
- Variations in radar image conditions (e.g., sea state, sensor angle) affected generalization.

2.1.3 Implementation Machine Learning for the Detection of Oil Spills (Kubat, Holte, & Matwin)

Architecture/Approach:

1. **Dataset Collection:** The dataset consists of Synthetic Aperture Radar (SAR) images obtained from satellite sources and environmental monitoring platforms. It includes two main classes: oil spill regions and non-oil regions (lookalikes such as low wind areas or natural films).

2. **Preprocessing:**
   The preprocessing steps are performed to enhance image quality and prepare data for analysis:

   I. **Noise Reduction** – Speckle noise inherent in SAR images is reduced using filtering techniques to improve clarity.

   II. **Normalization** – Pixel intensity values are standardized to maintain consistency across different images.

   III. **Segmentation** – Images are divided into regions of interest (ROI) to isolate potential oil spill areas.

   IV. **Feature Extraction** – Important features such as brightness, area, shape, and texture are extracted from each segmented region.

3. **Feature Engineering and Dimensional Handling:**

   • The extracted features are converted into structured feature vectors representing each region.

   • These vectors capture distinguishing characteristics between oil spills and lookalikes.

   • Feature selection techniques are applied to retain the most relevant attributes, improving model efficiency and reducing computational complexity.

4. **Model Training:**

   • **Decision Tree (C4.5):** Learns hierarchical classification rules based on feature thresholds (e.g., brightness, shape).

   • **1-Nearest Neighbor (1-NN):** Classifies regions by comparing them to the closest training sample using distance metrics like Euclidean distance.

• **SHRINK Algorithm:** Adjusts classification bias and thresholds to handle class imbalance, as oil spills are rare compared to non-oil regions.

5. **Evaluation and Testing:**

• The trained models are evaluated on unseen SAR images.

• Performance is measured using metrics such as accuracy, precision, recall, and confusion matrix.

• Special focus is given to minimizing false positives (lookalikes classified as oil) and false negatives (missed oil spills).

6. **Deployment:**

• The best-performing model is integrated into the CEHDS (Coastal Environmental Hazard Detection System).

• The system provides real-time oil spill detection and supports experts in monitoring and decision-making.

• The deployment ensures scalability, fast processing, and ease of use through a simple interface.

2.2. Oil Spill Identification from Satellite Images Using Deep Neural Networks (Krestenitis et al.)

2.2.1 Introduction

The paper *"Oil Spill Identification from Satellite Images Using Deep Neural Networks"* by Krestenitis et al. explores the application of deep learning for automatic oil spill detection in synthetic aperture radar (SAR) imagery. Unlike classical machine learning methods that rely on hand-crafted features, this approach employs Convolutional Neural Networks (CNNs) to learn features directly from raw pixel data, achieving pixel-level precision through semantic segmentation.

The study evaluates several advanced architectures:

- U-Net: An encoder-decoder CNN with skip connections, effective for biomedical and remote-sensing segmentation tasks.
- LinkNet: An encoder-decoder model using a ResNet backbone, optimized for efficiency and preserving spatial details.
- DeepLabv3+: An advanced segmentation framework with Atrous Spatial Pyramid Pooling (ASPP) and an enhanced decoder for sharper object boundaries.

All implementations used Keras (for *U-Net*, *LinkNet*, *DeepLabv3+*) and TensorFlow (for *DeepLabv2*), focusing on pixel-wise classification of "oil spill" and "non-oil" regions for precise contamination mapping. The method enhances detection accuracy by integrating local texture with global spatial features across marine areas.Among the models, U-Net achieved the best accuracy, approximately 53.79%, demonstrating its strength in fine-grained segmentation despite the challenging visual characteristics of radar data.

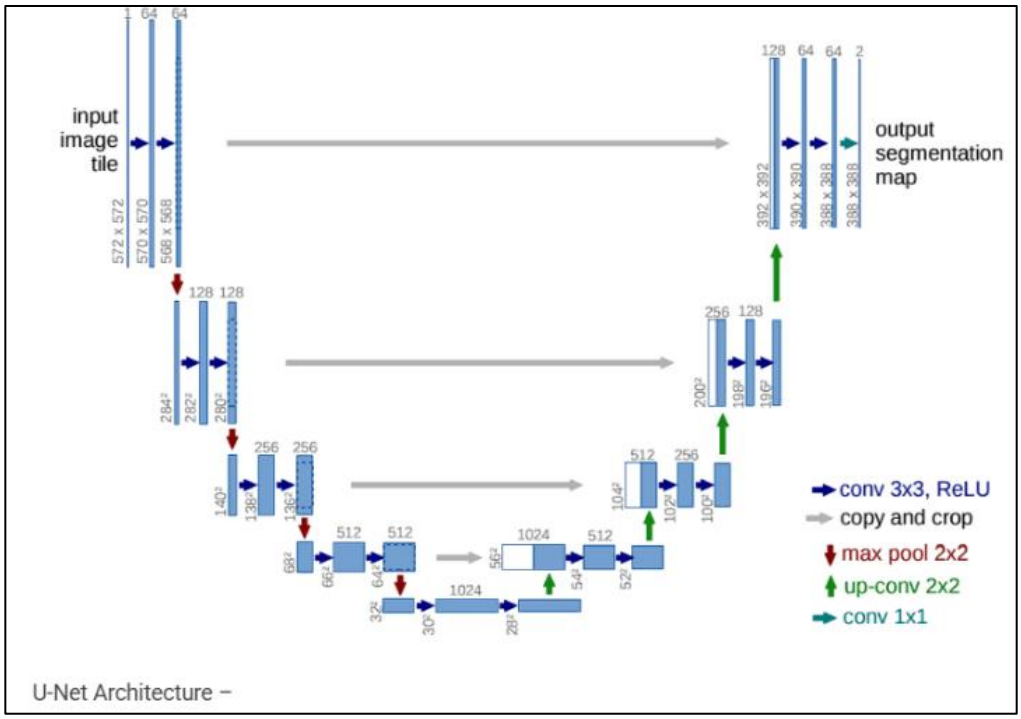

Figure 2.2.1 U-Net Architecture

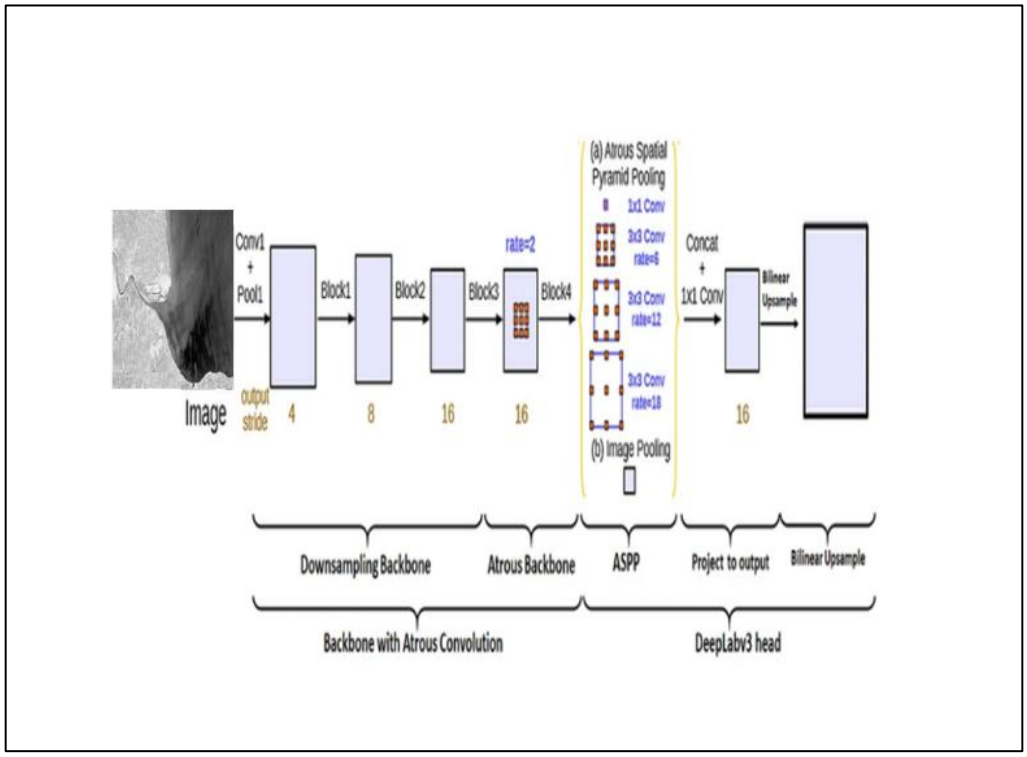

Figure 2.2.2 LinkNet

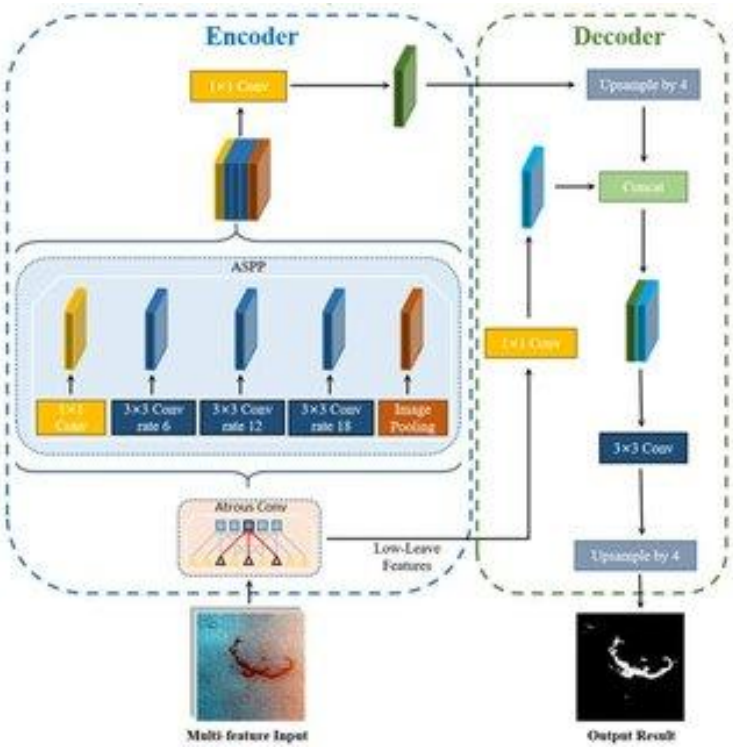

Figure 2.2.3 DeepLabv3+ Architecture

2.2.2 Merits, Demerits and Challenges

Merits:

- **Automatic feature learning:** CNNs eliminate the need for manual feature extraction, learning spatial and textural patterns directly from images.
- **High spatial accuracy:** Pixel-wise labeling provides detailed and localized detection of oil spills.
- **Robust architecture design:** Encoder-decoder structures and skip connections help preserve context and boundary details.
- **Transfer learning capability:** Pretrained backbones (e.g., ResNet, Xception) improve performance even with limited labeled datasets.
- **DeepLabv3+ innovation:** ASPP captures multi-scale contextual information, improving segmentation of irregular oil shapes.

Demerits:

- **Computationally intensive:** Deep networks require high processing power (GPUs) and long training times.
- **Large dataset requirement:** Deep models generally perform best with large annotated datasets, which are often scarce for oil spill imagery.
- **Moderate accuracy:** Although improved over classical methods, performance (≈ 53.79%) still leaves room for enhancement.
- **Overfitting Risk:** Deep models may memorize training data instead of generalizing well, especially when the dataset is limited.
- • **Sensitivity to Noise:** SAR images contain speckle noise, which can negatively affect feature extraction and model accuracy.

Challenges:

- **Data Scarcity**
  - Limited availability of labeled **SAR oil spill images** restricts effective deep learning model training.
  - Annotation of SAR images requires **expert knowledge**, making the labeling process time-consuming and expensive.
  - Small datasets reduce model robustness and increase the risk of **overfitting**. .
- **Class Imbalance**
  - Oil spill pixels occupy only a **small portion** of the image compared to the background.
  - Models tend to become **biased toward the majority class (non-oil regions)**.
  - This reduces the model's ability to accurately detect oil spill regions.
  - Techniques like **resampling and weighted loss functions** are required to handle imbalance effectively.
- **Generalization Issues**
  - Models may not generalize well across different sensors or environmental conditions.
  - Variations such as wind, waves, and sensor types affect image characteristics.
  - Models trained on one dataset may perform poorly on unseen or real-world data**.**

2.2.3 Implementation of Oil Spill Identification from Satellite Images Using Deep Neural Networks

The implementation follows a **deep learning–based semantic segmentation pipeline**, structured as follows:

Data Preparation and Preprocessing:

- SAR satellite images are preprocessed to reduce speckle noise and enhance contrast.
- Ground-truth masks (oil vs. non-oil) are generated for supervised learning.
- Data augmentation techniques (rotation, flipping, scaling) improve model robustness.

Model Architecture:

- **U-Net:**
  - **Encoder:** Extracts hierarchical features via convolution and pooling operations, compressing the image into high-level representations.
  - **Decoder:** Upsamples the feature maps, using skip connections to combine low-level spatial information from the encoder, producing detailed segmentation maps.
- **LinkNet:**
  - Employs a **ResNet encoder** with residual blocks to capture deeper semantic features while mitigating vanishing gradients.
  - The decoder mirrors the encoder, ensuring spatial precision with fewer parameters.
- **DeepLabv3+:**
  - **Encoder:** Utilizes **Atrous Spatial Pyramid Pooling (ASPP)** with multiple dilation rates (6, 12, 18) to capture multi-scale context.
  - **Decoder:** Merges encoder outputs with low-level features from early layers to recover sharp object boundaries.

2.3 Custom Convolutional Neural Network with Lee Filter Preprocessing

2.3.1 Introduction

The paper *"A Deep Learning-Based Oil Spill Detection Approach Using SAR Remote Sensing Images"* presents an efficient and automated solution for detecting oil spills in marine environments using advanced deep learning techniques. Oil spill detection is a critical task for environmental monitoring, as timely identification helps minimize ecological damage and supports effective response strategies. Synthetic Aperture Radar (SAR) imagery is widely used for this purpose due to its capability to capture images under all weather conditions and during both day and night. However, SAR images are inherently affected by speckle noise, which makes accurate detection a challenging problem.

To overcome these limitations, the proposed approach introduces a **custom Convolutional Neural Network (CNN)** specifically designed to handle SAR image characteristics. Unlike traditional machine learning methods that depend on handcrafted features or complex segmentation architectures, this model focuses on a lightweight and computationally efficient CNN classifier. This reduces model complexity while still maintaining reliable detection performance.A key component of this approach is the integration of the **Lee Filter** as a preprocessing step. The Lee Filter effectively reduces speckle noise while preserving important structural and textural information in the image. This enhancement improves the visibility of oil spill regions and ensures that meaningful features are retained for further processing.

The CNN architecture is composed of multiple layers, including convolutional layers for feature extraction, ReLU activation functions for introducing non-linearity, pooling layers for reducing dimensionality, and fully connected layers for classification. Finally, a Softmax layer is used to generate probability-based outputs for decision-making.Overall, this approach provides a robust, scalable, and end-to-end framework for oil spill detection. By combining effective preprocessing with a tailored CNN architecture, the method reduces reliance on manual feature engineering and offers a practical solution for real-world SAR-based environmental monitoring systems.

- **Dataset Collection**
  - The dataset consists of **ENVISAT ASAR SAR images** obtained from remote sensing sources.
  - It contains two classes: **Oil Spill regions** and **Non-Oil regions (lookalikes)**.
  - The dataset includes variations in environmental conditions such as wind, waves, and sensor angles.

- **Preprocessing**
  - The preprocessing stage improves SAR image quality and prepares it for CNN input:
  - **Lee Filter (Noise Reduction):**
    The Lee Filter is applied to reduce speckle noise while preserving edges and important texture features.
  - **Normalization:**
    Pixel values are scaled to a uniform range for consistent model performance.
  - **Resizing:**
    Images are resized into fixed dimensions suitable for CNN input.
  - **Patch Extraction:**
    Images are divided into smaller regions (patches) for localized classification.

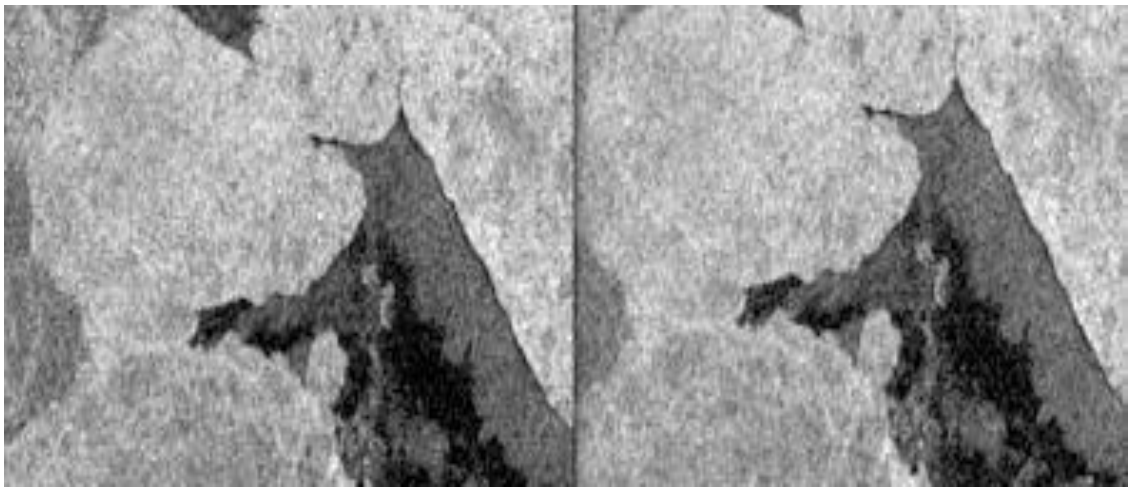

Figure 2.3.1 Lee Filter Processing

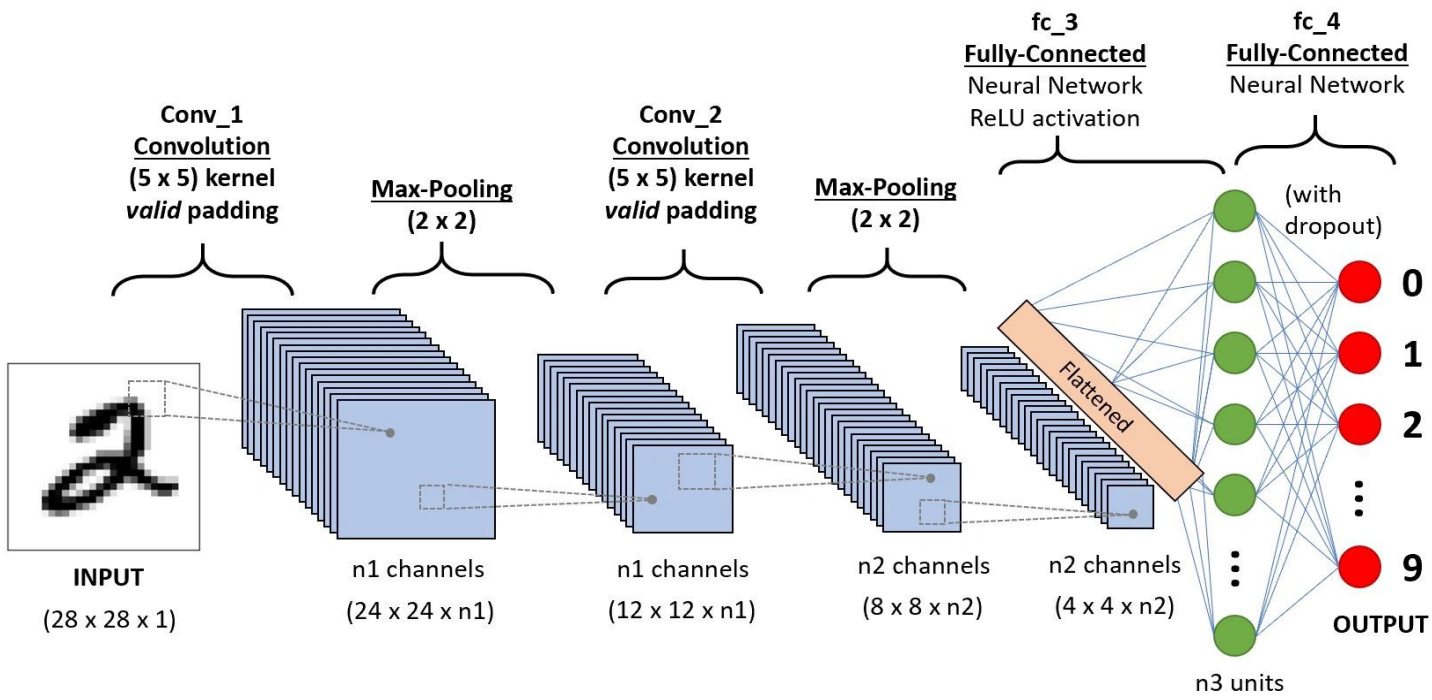

Figure 2.3.2 Convolutional Neural Network

2.3.2 Merits, Demerits and Challenges

Merits:

1. The process effectively estimates leaf age by analyzing the leaf area and colour intensity, which helps determine the crop's growth stage and maturity level.
2. Preprocessing steps like Otsu's segmentation and binarization remove background noise and improve image clarity.
3. Compared to complex deep learning models, K-Means requires less computational power and memory.
4. The algorithm accurately divides the leaf into distinct color-based clusters, enabling clear identification of healthy, partially infected, and severely diseased regions.
5. The system reduces manual labor and human dependency in disease identification.

Demerits:

1. The algorithm requires the number of clusters to be predefined; choosing an incorrect K value can lead to inaccurate segmentation or misclassification of regions.
2. When leaf images have uneven lighting, shadows, or overlapping regions, K-Means may fail to correctly separate diseased areas.
3. K-Means performs poorly with irregularly shaped disease patches that do not conform to well-defined boundaries.
4. The method relies primarily on color intensity for clustering and ignores important texture and shape information that could improve accuracy.

Challenges:

1. Removing background elements completely is crucial; residual noise or unwanted areas can mislead the clustering process.
2. Similar color tones between healthy green regions and early-stage disease spots can cause misclassification, reducing accuracy.
3. Processing large numbers of high-resolution images or integrating the system for real-time field applications can increase computational demand.

2.3.3 Implementation of Custom Convolutional Neural Network with Lee Filter Preprocessing

The process is divided into two main parts:

1. Oil Spill Region Detection (Segmentation)
2. Oil Spill Classification and Verification

1. Oil Spill Region Detection (Segmentation)

Steps:

• Acquire **SAR (Synthetic Aperture Radar) image** of the ocean surface.

• Apply **Lee Filter** to remove multiplicative noise while preserving important features.

• Perform **image enhancement** to improve visibility (oil spills appear as dark patches).

• Resize and normalize the image for model compatibility.

• Input the preprocessed image into **deep learning segmentation models**:

- U-Net
- LinkNet
- DeepLabv3+

  • Each model performs **pixel-level segmentation** to detect potential oil spill regions.

  • Combine outputs using an **ensemble approach** to improve accuracy and robustness.

  • Generate a final **oil spill mask highlighting suspected regions**.

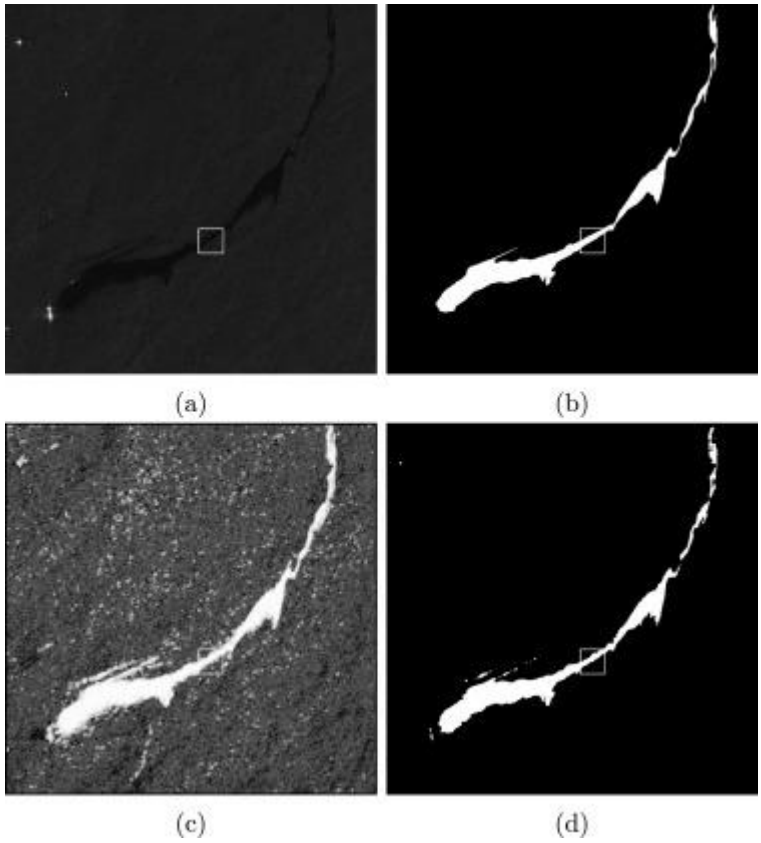

Fig 2.3.3: Oil Spill Segmentation Output

2. **Oil Spill Classification and Verification**

Steps:

• Take segmented oil spill regions as input.

• Extract **region-based features** from detected areas:

- Texture features (GLCM: contrast, homogeneity, energy)

- Intensity features (mean, variance, skewness)
- Shape features (area, perimeter, irregularity)

• Convert pixel-level data into **feature vectors**.

• Apply **machine learning classifiers** to verify detected regions:

- Random Forest
- K-Nearest Neighbors (KNN)

• Classify each region as:

- Oil Spill
- Look-alike (false detection such as low-wind zones, algae, etc.)

• Remove false positives and retain only **confirmed oil spill regions**.

• Display final output with **accurate oil spill mapping and classification results**.

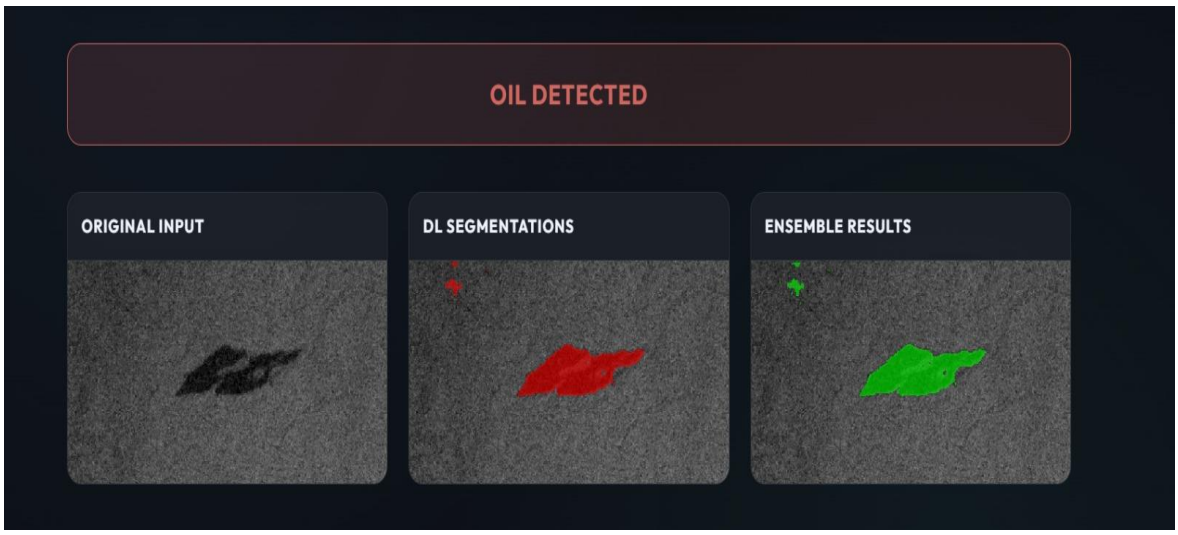

Fig 2.3.4: Oil Spill Classification and Verification

Machine Learning Classification (Random Forest / KNN)

• Machine learning algorithms are used as a **verification layer** to improve reliability.

• They classify segmented regions based on extracted features.

Working:

- Input feature vectors obtained from segmented regions.

- For **Random Forest**:
- For **K-Means Clustering**

Random Forest Classification:

- Random Forest is a **supervised machine learning algorithm** used for classification and regression tasks.
- It is an **ensemble technique** that combines multiple decision trees to improve accuracy.
- Randomly select subsets of data using **bootstrap sampling** to train each decision tree
- For each tree, a random subset of features is chosen to build the model.
- Each decision tree makes its own prediction for the given input data.
- The final output is determined by **majority voting**, improving accuracy and reducing overfitting.
- It is **robust to noise and outliers**, making it suitable for real-world data like SAR images.
- It provides **better generalization** compared to a single decision tree.
- It can handle **large datasets and high-dimensional features** efficiently.

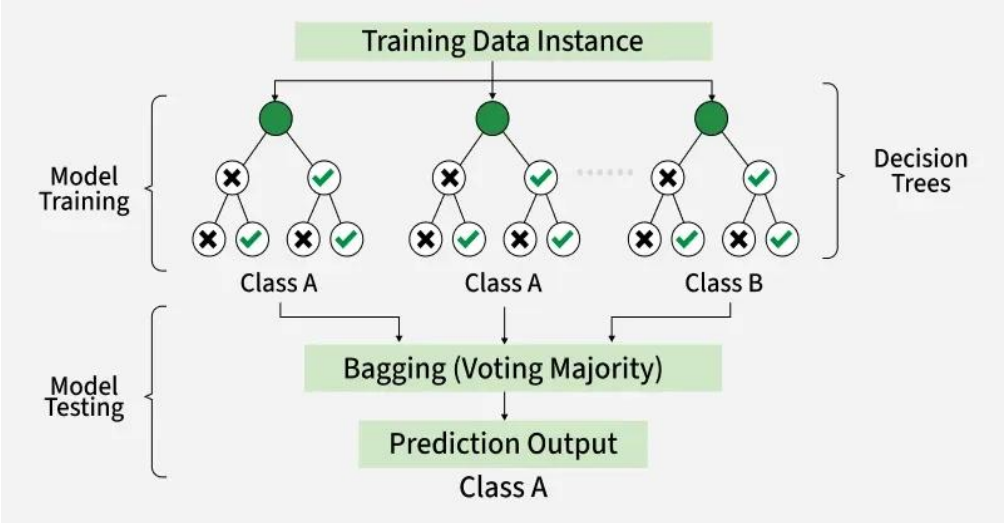

Fig 2.3.5: Random Forest Classification:

K-Nearest Neighbors (KNN):

- KNN is a supervised machine learning algorithm used for classification and regression.
- It classifies a data point based on the majority class among its K nearest neighbors.

Working:

- Choose the value of K (number of neighbors).
- Calculate the distance between the test data point and all training data points (commonly using Euclidean distance).
- Select the K nearest data points.
- Assign the class based on majority voting among those neighbors.

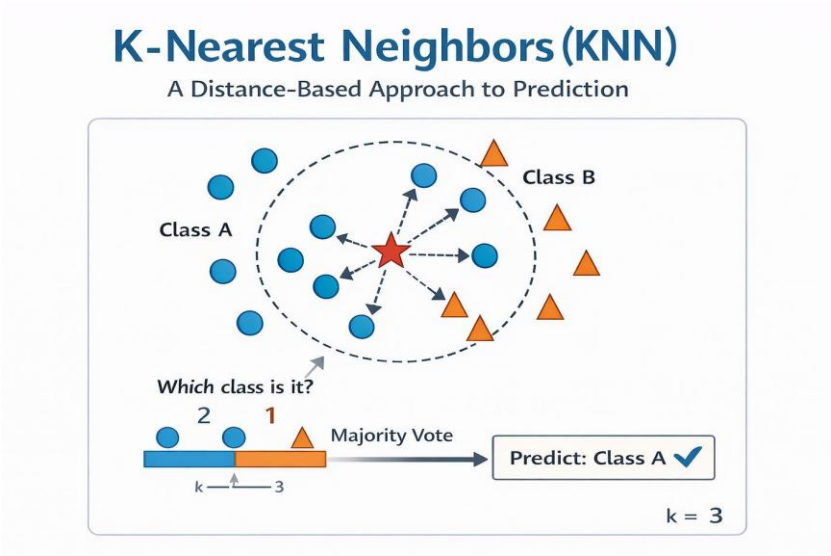

Fig 2.3.6: K-Nearest Neighbors

CHAPTER 3
PROPOSED SYSTEM

CHAPTER 3

PROPOSED SYSTEM

The proposed system introduces a **multi-stage hybrid framework** for precise oil spill detection and classification using Synthetic Aperture Radar (SAR) imagery. The system is designed to overcome the limitations of single-stage models by combining the strengths of both deep learning and traditional machine learning techniques. Initially, SAR images are collected and preprocessed to improve image quality and remove noise. Since SAR images are affected by multiplicative noise, adaptive filtering techniques such as **Lee filtering** are applied to preserve important structural features while reducing noise.

Image enhancement techniques are also used to improve visibility, as oil spills typically appear as dark patches on the ocean surface. In the next stage, the preprocessed images are passed through an **ensemble of deep learning-based semantic segmentation models**, including **U-Net, LinkNet, and DeepLabv3+**. These models work together to accurately identify potential oil spill regions at the pixel level. The ensemble approach improves robustness, reduces model bias, and enhances detection accuracy under varying ocean conditions.

Once the oil spill regions are segmented, a feature extraction module is applied to convert pixel-level outputs into meaningful regional features. These include **texture features (GLCM-based), intensity features (mean, variance, skewness), and shape features (area, perimeter, irregularity)**, which help distinguish oil spills from look-alike phenomena.

In the final stage, **machine learning classifiers such as Random Forest and K-Nearest Neighbors (KNN)** are used to verify the detected regions. This step acts as a validation layer to eliminate false positives caused by look-alikes like low-wind zones, algal blooms, or calm water surfaces.This multi-stage pipeline ensures accurate, reliable, and interpretable detection of oil spills, enabling effective environmental monitoring and timely response to marine pollution.

3.1. Objective of Proposed Model

- To collect a dataset of SAR images containing both oil spill and non-oil spill regions.
- To preprocess SAR images using noise reduction (Lee filtering) and enhancement techniques.
- To apply deep learning-based semantic segmentation models (U-Net, LinkNet, DeepLabv3+) for accurate oil spill region detection.
- To combine multiple segmentation models using an ensemble approach to improve robustness and reduce false detections.
- To extract discriminative features such as texture, intensity, and shape from segmented regions.
- To use machine learning classifiers (Random Forest, KNN) for validation and reduction of false positives.
- To evaluate model performance using metrics such as accuracy, precision, recall, and F1-score.
- To ensure reliable detection under varying ocean and environmental conditions.

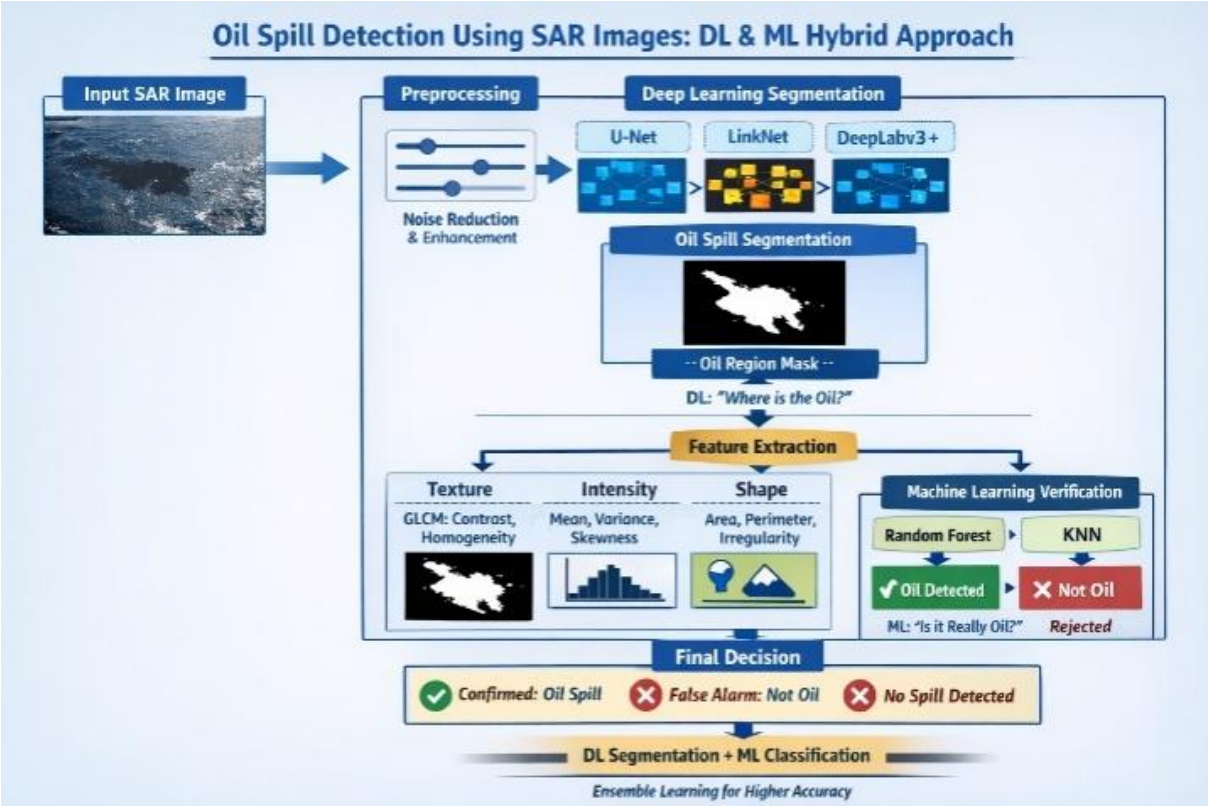

Fig 3.1.1: System Architecture

3.2. Algorithms Used for Proposed Model

The proposed system integrates both image processing and advanced learning algorithms to achieve high accuracy in oil spill detection.

- **Image Preprocessing Techniques**

  Preprocessing improves SAR image quality by reducing noise and enhancing important features. Techniques such as Lee filtering are used to remove multiplicative noise while preserving edges and structural details. Image enhancement improves the visibility of oil spill regions.

- **Semantic Segmentation Models (U-Net, LinkNet, DeepLabv3+)**

  These deep learning models perform pixel-level classification to identify oil spill regions.

  - U-Net provides strong localization with skip connections.
  - LinkNet offers computational efficiency for faster processing.
  - DeepLabv3+ captures multi-scale contextual information using atrous convolutions.

    The ensemble of these models improves segmentation accuracy and robustness.

- **Ensemble Learning**

  Outputs from multiple segmentation models are combined to generate a final oil spill mask. This reduces individual model bias and improves generalization across different environmental conditions.

- **Feature Extraction (GLCM, Intensity, Shape Features)**

  After segmentation, features are extracted to describe oil spill characteristics:

  - Texture features using Gray Level Co-occurrence Matrix (GLCM)
  - Intensity features such as mean and variance
  - Shape features like area, perimeter, and irregularity

- **Machine Learning Classifiers (Random Forest, KNN)**

  These classifiers validate segmented regions and eliminate false positives.

  - Random Forest improves robustness using multiple decision trees.
  - KNN classifies based on similarity using distance measures.

- **Deep Learning (CNN-based Models)**

  CNN-based architectures automatically learn spatial and contextual features from SAR images, enabling accurate detection of complex oil spill patterns.

- **Evaluation Metrics**

  Performance is measured using:

  - Accuracy
  - Precision
  - Recall
  - F1-score

  These metrics ensure the reliability and effectiveness of the model.

3.3. Designing

The system is designed as a **multi-stage, modular pipeline** to ensure efficiency, scalability, and real-world usability in marine environmental monitoring.

The application follows a structured architecture consisting of **image acquisition, preprocessing, segmentation, feature extraction, and classification stages**. Each module is designed independently, making the system easy to maintain and extend.The workflow begins with SAR image input, which is processed through noise reduction and enhancement techniques. The cleaned images are then passed to an ensemble of deep learning models (U-Net, LinkNet, DeepLabv3+) for segmentation. The resulting oil spill regions are converted into feature representations using texture, intensity, and shape descriptors.

These features are then fed into machine learning classifiers for validation, ensuring that only true oil spill regions are retained while eliminating look-alike patterns. This hybrid design significantly reduces false alarms and improves detection reliability.The system can be implemented as a **web-based application**, where users can upload SAR images and receive results in real-time. The backend processes the image through the pipeline, while the frontend displays the segmented output, classification results, and additional insights.

The result visualization module enhances usability by presenting:

- Detected oil spill regions (highlighted images)
- Classification results (oil spill / non-oil spill)
- Confidence scores
- Analytical insights for decision-making

This design ensures that the system is not only technically robust but also highly practical for real-world deployment in **oil spill monitoring, marine environmental protection, and rapid disaster response scenarios**.

3.3.1. UML Diagrams

The UML diagrams of the proposed system illustrate the structure, workflow, and interaction of components involved in the multi-stage oil spill detection pipeline. These diagrams provide a clear understanding of how data flows through different stages such as preprocessing, deep learning-based segmentation, feature extraction, and machine learning classification. They help in visualizing both the functional behavior and system architecture, making the design easier to analyze, implement, and communicate.

1. Use Case Diagram

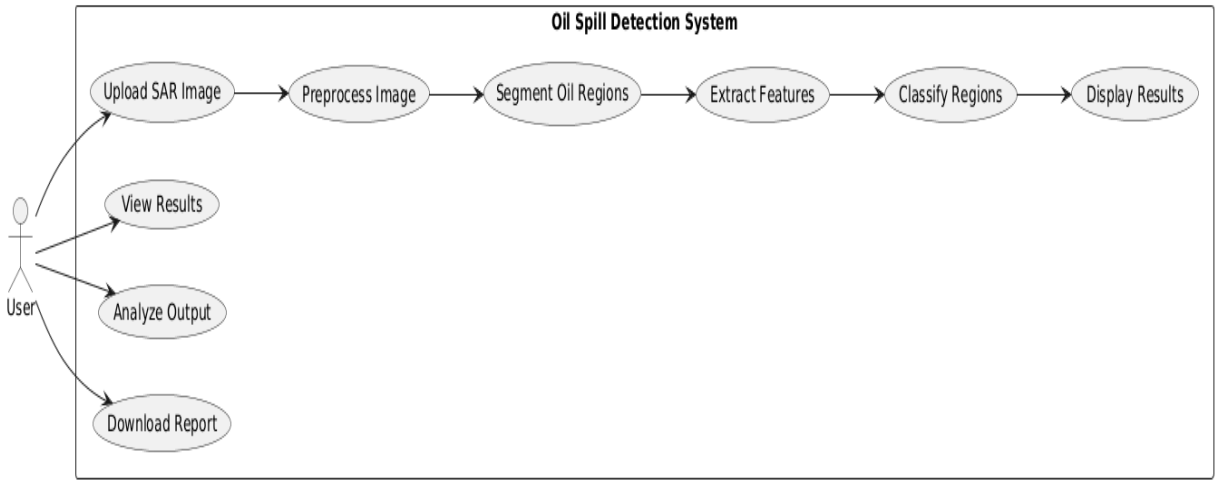

Fig 3.3.1: Use Case Diagram

The above diagram represents the Use Case Diagram of the Oil Spill Detection System, illustrating the interaction between the user and the system functionalities.

The process begins with the User (researcher/analyst), who accesses the system through a web-based interface. The user uploads a SAR image of the ocean surface for analysis. The system then performs multiple operations, including preprocessing, segmentation, feature extraction, and classification.The user can view the detected oil spill regions, analyze the results, and download reports for further use. Internally, the system processes the input image through various stages such as noise reduction, deep learning-based segmentation, and machine learning-based verification.This diagram highlights how the user interacts with the system and how the system provides functionalities for efficient oil spill detection and analysis.

2. Class Diagram

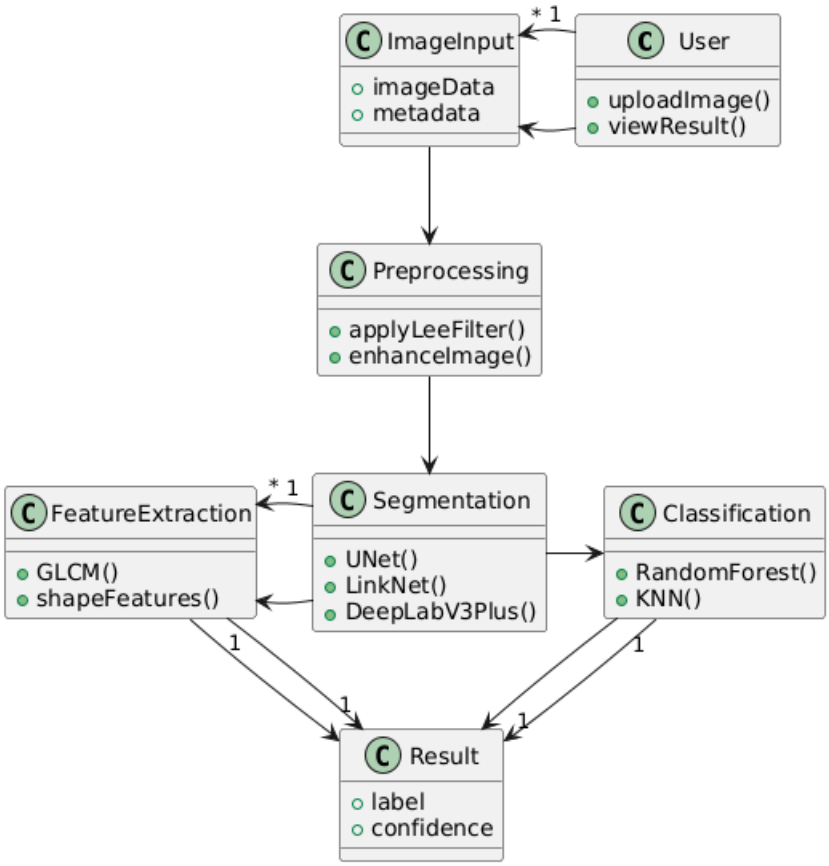

Fig 3.3.2: Class Diagram

The above diagram represents the Class Diagram of the Oil Spill Detection System, showing the structural design and relationships between different components.The system consists of several classes, including User, Image Input, Preprocessing, Segmentation, Feature Extraction, Classification, and Result. The User interacts with the system by uploading SAR images through the Image Input class.The Image Input is processed by the Preprocessing class, where noise reduction using Lee filtering and image enhancement are performed. The processed data is then passed to the Segmentation class, which uses deep learning models such as U-Net, LinkNet, and DeepLabv3+ to detect potential oil spill regions.

The segmented regions are forwarded to the Feature Extraction class, where important features like texture, intensity, and shape are extracted. These features are then used by the Classification class, which applies machine learning algorithms such as Random Forest and KNN to verify whether the detected regions are actual oil spills or look-alikes. Finally, the Result class stores and displays the classification output, including labels and confidence scores. The relationships between classes, along with multiplicities, represent the flow of data and dependencies within the system.

3. Sequence diagram

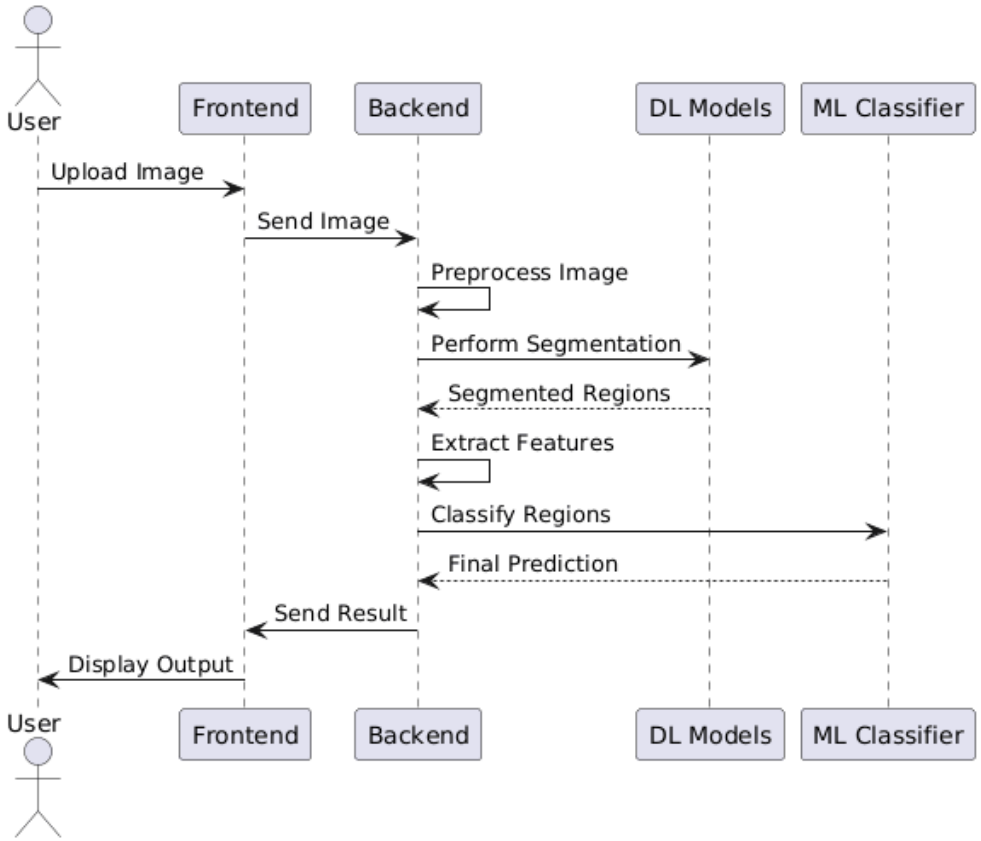

Fig 3.3.3: Sequence Diagram

The above diagram represents the Sequence Diagram of the Oil Spill Detection System, showing the step-by-step interaction between system components over time.The process begins with the User, who uploads a SAR image through the frontend interface. The frontend sends the image to the Backend (Flask) for processing. The backend first performs preprocessing operations such as noise reduction and image enhancement.The processed image is then forwarded to the Deep

Learning Models, where segmentation is performed using U-Net, LinkNet, and DeepLabv3+. These models generate segmented regions that potentially contain oil spills.Next, the backend extracts features from these regions and sends them to the Machine Learning Classifier, such as Random Forest or KNN. The classifier verifies the regions and removes false positives caused by look-alike patterns.Finally, the classification results are returned to the backend and displayed to the user through the frontend interface, showing detected oil spill regions and prediction outcomes.

4. Activity Diagram

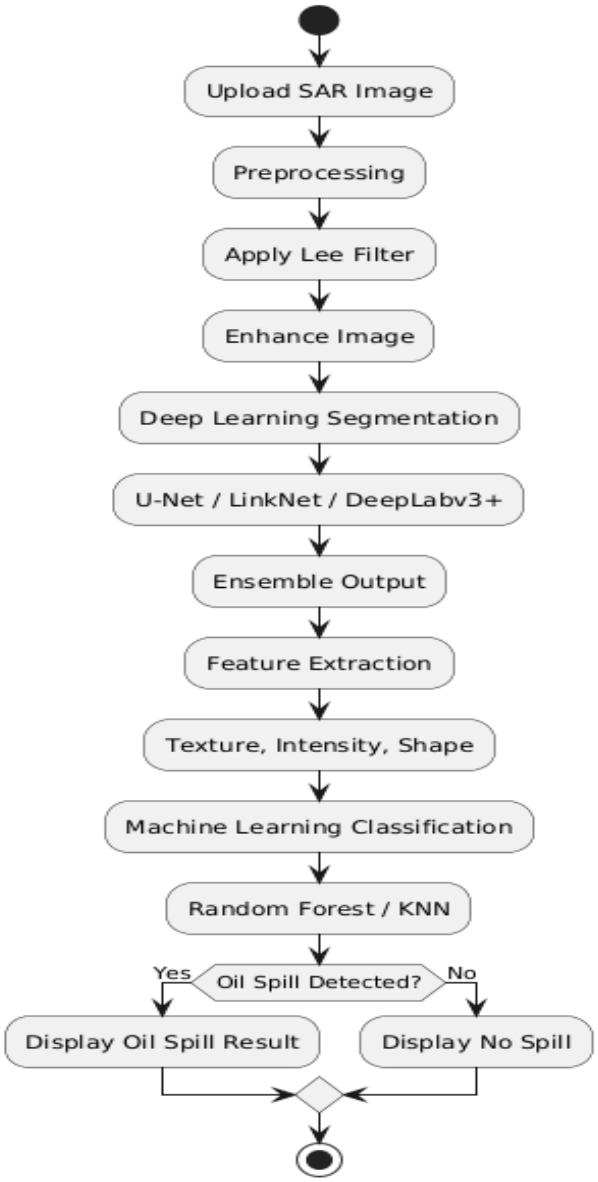

Fig 3.3.4: Activity Diagram

The above diagram represents the **Activity Diagram** of the Oil Spill Detection System, illustrating the workflow of the entire process.The process starts with the user uploading a SAR image. The system then performs preprocessing, which includes noise reduction using Lee filtering and image enhancement.After preprocessing, the image is passed through deep learning-based segmentation models such as U-Net, LinkNet, and DeepLabv3+, which identify potential oil spill regions. The outputs from these models are combined using an ensemble approach to improve accuracy.Next, feature extraction is performed to obtain texture, intensity, and shape features from the segmented regions.

These features are then used by machine learning classifiers like Random Forest or KNN to verify the results.A decision is made based on classification: if an oil spill is detected, the system displays the result; otherwise, it indicates no oil spill. The process ends with the final output being shown to the user.This diagram clearly represents the sequential flow of activities involved in detecting and classifying oil spills.

3.4. Stepwise Implementation and Code snippets

PREPROCESSING CODE (FOR ML + DL):

```
import os
import cv2
import numpy as np

IMG_SIZE = 128
DATASET_PATH = "dataset/"

images = []
masks = []   # for DL segmentation
labels = []  # for ML classification

def load_data():
    categories = ["no_oil", "oil_spill"]

    for label, category in enumerate(categories):
        img_folder = os.path.join(DATASET_PATH, category, "images")
        mask_folder = os.path.join(DATASET_PATH, category, "masks")

        for file in os.listdir(img_folder):
```

```
        img_path = os.path.join(img_folder, file)
        mask_path = os.path.join(mask_folder, file)

        try:
            img = cv2.imread(img_path)
            img = cv2.resize(img, (IMG_SIZE, IMG_SIZE))

            # Normalize
            img_norm = img / 255.0

            # Load mask (for DL)
            mask = cv2.imread(mask_path, 0)
            mask = cv2.resize(mask, (IMG_SIZE, IMG_SIZE))
            mask = mask / 255.0

            images.append(img_norm)
            masks.append(mask)
            labels.append(label)

        except:
            pass

load_data()

images = np.array(images)
masks = np.array(masks)
labels = np.array(labels)

# ML format (flatten)
data_ml = images.reshape(len(images), -1)

print("Data Loaded:", images.shape)
```

ML TRAINING (RANDOM FOREST + KNN):

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
import joblib

X_train, X_test, y_train, y_test = train_test_split(
    data_ml, labels, test_size=0.2, random_state=42
)
```

```
print("\nTraining Random Forest...")
rf_model = RandomForestClassifier(n_estimators=100)
rf_model.fit(X_train, y_train)

rf_pred = rf_model.predict(X_test)
print("RF Accuracy:", accuracy_score(y_test, rf_pred))

joblib.dump(rf_model, "rf_model.pkl")


# -------- KNN --------
print("\nTraining KNN...")
knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(X_train, y_train)

knn_pred = knn_model.predict(X_test)
print("KNN Accuracy:", accuracy_score(y_test, knn_pred))

joblib.dump(knn_model, "knn_model.pkl")
```

DL TRAINING (U-NET, LINKNET, DEEPLABV3+):

```
import segmentation_models as sm
from sklearn.model_selection import train_test_split

sm.set_framework('tf.keras')

X_train, X_test, y_train, y_test = train_test_split(
    images, masks, test_size=0.2, random_state=42
)

# Common parameters
BACKBONE = 'resnet34'
LR = 0.0001

# Compile function
def compile_model(model):
    model.compile(
        optimizer='adam',
        loss=sm.losses.bce_jaccard_loss,
        metrics=[sm.metrics.iou_score]
    )
    return model


# -------- U-NET --------
```

```
print("\nTraining U-Net...")
unet = sm.Unet(BACKBONE, encoder_weights='imagenet')
unet = compile_model(unet)

unet.fit(X_train, y_train, epochs=10, batch_size=8)
unet.save("unet_model.h5")


# -------- LINKNET --------
print("\nTraining LinkNet...")
linknet = sm.Linknet(BACKBONE, encoder_weights='imagenet')
linknet = compile_model(linknet)

linknet.fit(X_train, y_train, epochs=10, batch_size=8)
linknet.save("linknet_model.h5")


# -------- DEEPLABV3+ --------
print("\nTraining DeepLabV3+...")
deeplab = sm.DeepLabV3Plus(BACKBONE, encoder_weights='imagenet')
deeplab = compile_model(deeplab)

deeplab.fit(X_train, y_train, epochs=10, batch_size=8)
deeplab.save("deeplab_model.h5")
```

FINAL INTEGRATED TESTING CODE:

```
import cv2
import numpy as np
import joblib
from tensorflow.keras.models import load_model

IMG_SIZE = 128

# Load models
rf_model = joblib.load("rf_model.pkl")
knn_model = joblib.load("knn_model.pkl")

unet = load_model("unet_model.h5", compile=False)
linknet = load_model("linknet_model.h5", compile=False)
deeplab = load_model("deeplab_model.h5", compile=False)


def preprocess(image_path):
    img = cv2.imread(image_path)
    img = cv2.resize(img, (IMG_SIZE, IMG_SIZE))
```

```
    img_norm = img / 255.0

    # ML format
    img_ml = img_norm.reshape(1, -1)

    # DL format
    img_dl = np.expand_dims(img_norm, axis=0)

    return img_ml, img_dl


def predict(image_path):
    img_ml, img_dl = preprocess(image_path)

    # -------- ML Predictions --------
    rf_pred = rf_model.predict(img_ml)
    knn_pred = knn_model.predict(img_ml)

    # -------- DL Predictions (Segmentation) --------
    unet_mask = unet.predict(img_dl)
    linknet_mask = linknet.predict(img_dl)
    deeplab_mask = deeplab.predict(img_dl)

    print("\n===== FINAL RESULTS =====")

    print("Random Forest:", "Oil Spill" if rf_pred[0] == 1 else "No Oil")
    print("KNN:", "Oil Spill" if knn_pred[0] == 1 else "No Oil")

    print("U-Net: Segmentation Done")
    print("LinkNet: Segmentation Done")
    print("DeepLabV3+: Segmentation Done")


# Test image
predict("test.jpg")
```

Frontend Code Snippet:

```
import React, { useState } from "react";

function App() {
  const [file, setFile] = useState(null);
  const [result, setResult] = useState(null);
  const [preview, setPreview] = useState(null);
```

```
// Handle file selection
const handleFileChange = e) => {
  const selectedFile = e.target.files[0];
  setFile(selectedFile);

  // Preview image
  if (selectedFile) {
    setPreview(URL.createObjectURL(selectedFile));
  }
};

// Send image to backend
const handleUpload = async () => {
  if (!file) {
    alert("Please select an image first!");
    return;
  }

  const formData = new FormData();
  formData.append("file", file);

  try {
    const response = await fetch("http://localhost:8000/predict", {
      method: "POST",
      body: formData,
    });

    const data = await response.json();
    setResult(data);
  } catch (error) {
    console.error("Error:", error);
    alert("Error connecting to backend!");
  }
};

return (
  <div style={styles.container}>
    <h1 style={styles.title}> Oil Spill Detection System</h1>

    {/* Upload Section */}
    <div style={styles.uploadBox}>
      <input type="file" onChange={handleFileChange} />
      <button style={styles.button} onClick={handleUpload}>
        Predict
      </button>
    </div>
```

```
{/* Preview Image */}
{preview && (
  <div style={styles.previewBox}>
    <h3> Uploaded Image</h3>
    <img src={preview} alt="preview" style={styles.previewImg} />
  </div>
)}

{/* Results Section */}
{result && (
  <div style={styles.resultBox}>
    <h2> Results</h2>

    <p><strong>Random Forest:</strong> {result.rf}</p>
    <p><strong>KNN:</strong> {result.knn}</p>

    <h3> Segmentation Outputs</h3>

    <div style={styles.imageContainer}>
      <div>
        <p>U-Net</p>
        <img
          src={`${result.unet}`}
          alt="unet"
          style={styles.outputImg}
        />
      </div>

      <div>
        <p>LinkNet</p>
        <img
          src={`${result.linknet}`}
          alt="linknet"
          style={styles.outputImg}
        />
      </div>

      <div>
        <p>DeepLabV3+</p>
        <img
          src={`${result.deeplab}`}
          alt="deeplab"
          style={styles.outputImg}
        />
      </div>
    </div>
  </div>
```

```
      )}
    </div>
  );
}

// Simple inline styling
const styles = {
  container: {
    textAlign: "center",
    fontFamily: "Arial",
    padding: "20px",
  },
  title: {
    color: "#0a3d62",
  },
  uploadBox: {
    margin: "20px",
  },
  button: {
    marginLeft: "10px",
    padding: "10px 20px",
    backgroundColor: "#0984e3",
    color: "white",
    border: "none",
    cursor: "pointer",
    borderRadius: "5px",
  },
  previewBox: {
    marginTop: "20px",
  },
  previewImg: {
    width: "300px",
    borderRadius: "10px",
  },
  resultBox: {
    marginTop: "30px",
  },
  imageContainer: {
    display: "flex",
    justifyContent: "center",
    gap: "20px",
    flexWrap: "wrap",
  },

export default App;
```

CHAPTER 4
RESULTS AND DISCUSSION

CHAPTER 4

4.1 Results and Discussion

Fig 4.1.1: Graphical User Interface for Uploading the Image

The developed system includes a user-friendly **Graphical User Interface (GUI)** that enables easy interaction with the oil spill detection model. The interface is designed using modern frontend technologies such as **Tailwind CSS**, providing a clean, responsive, and intuitive layout.The GUI allows users to upload **SAR (Synthetic Aperture Radar) images** through a simple file upload option, making it accessible even for non-technical users. The interface clearly displays the purpose of the system, guiding users in performing oil spill detection.Once the SAR image is uploaded, it is sent to the backend system implemented using **Flask**, where it undergoes preprocessing, segmentation, and classification

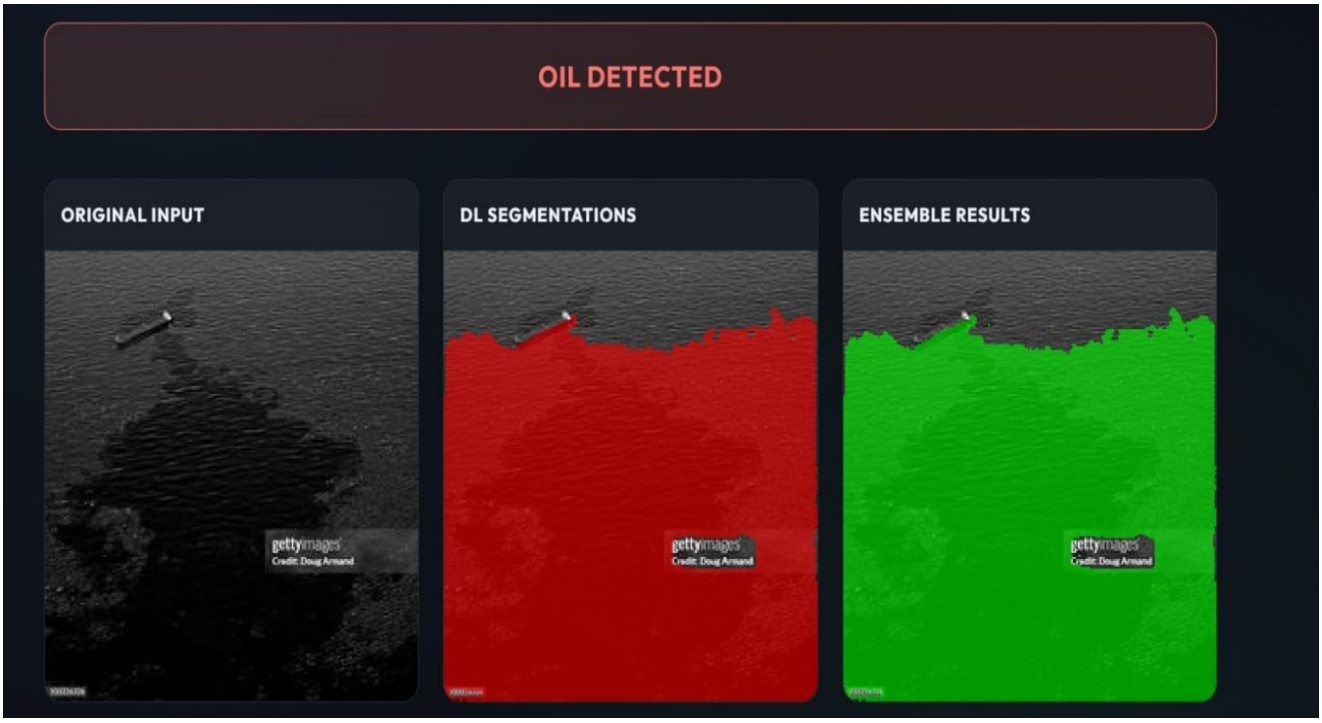

Fig 4.1.2: Oil spill Detection image

The above image demonstrates an automated pipeline for marine oil spill detection, showcasing the progression from raw data to refined analytical output. The **Original Input** displays grayscale aerial or satellite imagery where a vessel is seen trailing a large, low-reflectance dark patch, which is characteristic of an oil slick on the ocean surface. To identify this feature, a **Deep Learning (DL) Segmentation** model is applied, as seen in the center frame. This stage uses neural networks to generate a preliminary red mask over the detected anomaly; however, while highly sensitive, individual DL models can sometimes struggle with "noise" or imprecise boundaries in varying sea states.

The final **Ensemble Results** represent a more robust and accurate classification. by combining the outputs of multiple algorithms or models, the system filters out potential false positives and produces a cohesive green mask that more reliably defines the spill's true extent. This multi-stage approach is critical for environmental monitoring, as the resulting high-confidence data allows response teams to accurately calculate the surface area of the spill and model its drift patterns for more effective containment and cleanup efforts.

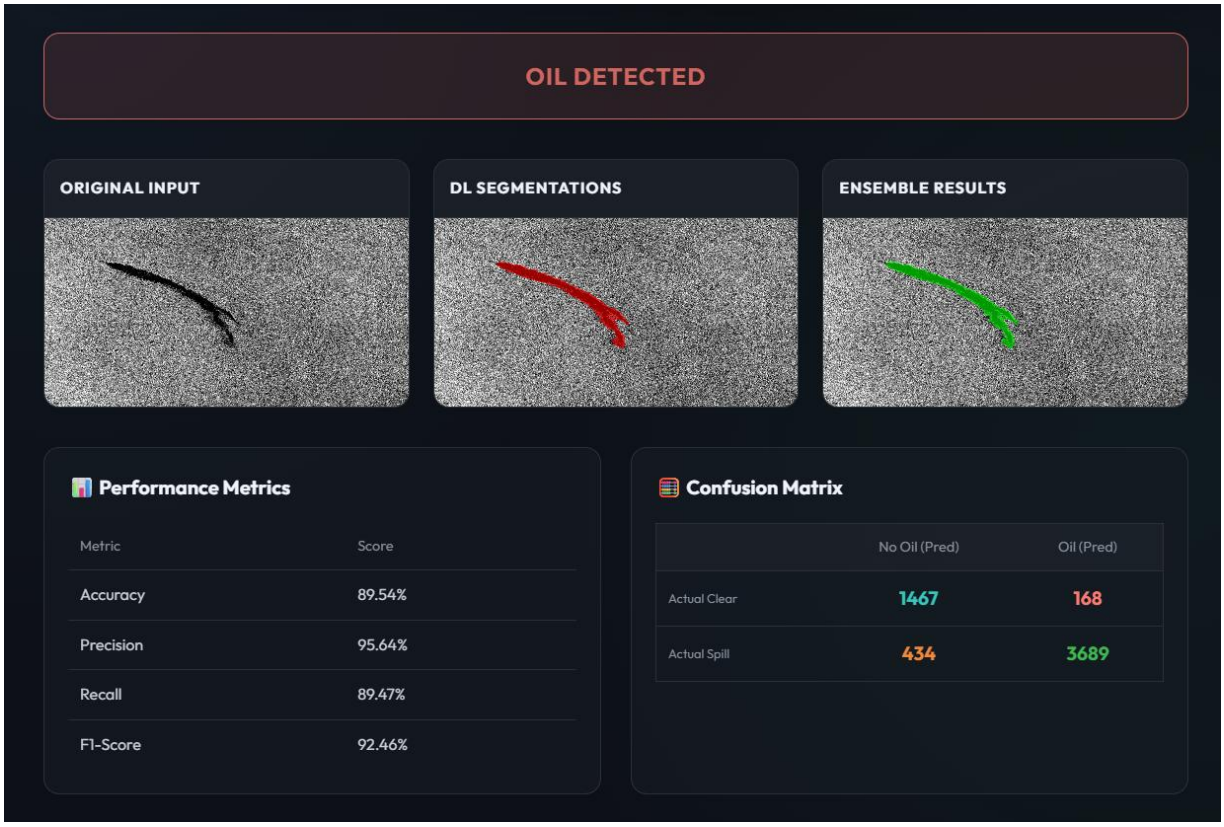

Fig 4.1.3: Oilspill Output of Uploaded image

The above interface represents the result screen of the Oil Spill Detection System after analysing a satellite or SAR image. A clear alert message “OIL DETECTED” indicates the presence main panels: “Original Input” shows the uploaded image, “DL Segmentations”highlights detected regions using deep learning (in red), and “Ensemble Results” presents the refined detection output (in green) for improved accuracy.Below, the “Performance Metrics” section summarizes the model’s effectiveness with Accuracy (89.54%), Precision (95.64%), Recall (89.47%), and F1-Score (92.46%). The “Confusion Matrix” further illustrates the comparison between actual and predicted results.of an oil spill, providing immediate understanding of the result.The interface displays three

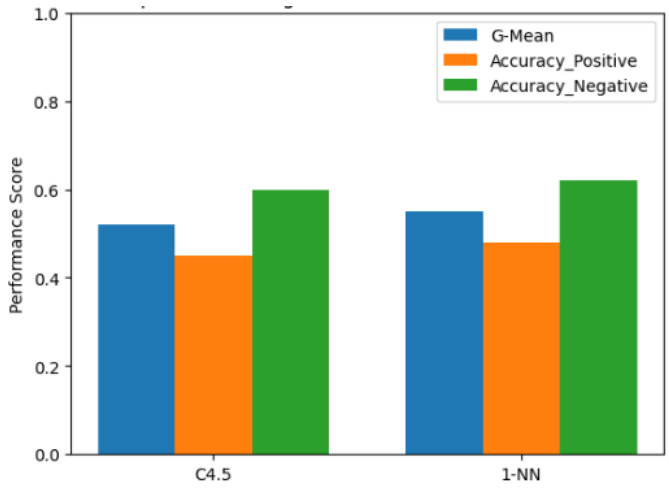

Fig 4.1.4: Performance Analysis of C4.5,1-NN

The bar chart compares the performance of two machine learning classifiers, C4.5 and 1-NN, across three key evaluation metrics: G-Mean, Accuracy_Positive, and Accuracy_Negative. For the C4.5 classifier, the G-Mean score is approximately 0.52, the Accuracy_Positive is around 0.45, and the Accuracy_Negative reaches about 0.60. In comparison, the 1-NN classifier shows slightly better results with a G-Mean of approximately 0.55, Accuracy_Positive of 0.50, and Accuracy_Negative of 0.64. Overall, the 1-NN algorithm outperforms the C4.5 decision tree across all three metrics, demonstrating more balanced and higher classification performance on both the positive and negative classes. The G-Mean, which is particularly useful for imbalanced datasets, indicates that 1-NN achieves a better trade-off between sensitivity and specificity. This suggests that 1-NN is the more effective model in this evaluation scenario.

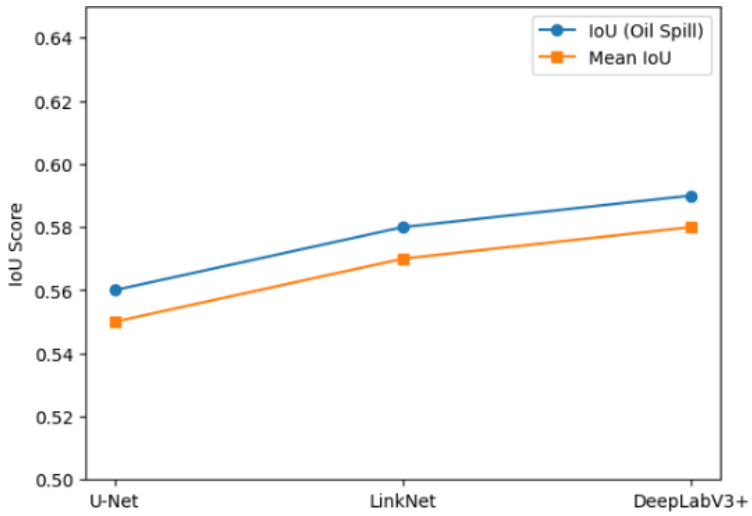

Fig 4.1.5: Performance Analysis of U-NET,Linknet,DeepLabV3+)

The provided graph presents a performance comparison of three semantic segmentation architectures U-Net, LinkNet, and DeepLabV3+ for oil spill detection using Intersection over Union (IoU) as the evaluation metric. The blue line represents the IoU specifically for the oil spill class, while the orange line shows the Mean IoU across all classes. U-Net achieves the lowest scores with an oil spill IoU of approximately 0.56 and a Mean IoU of 0.55. LinkNet performs moderately better, recording an oil spill IoU of around 0.585 and a Mean IoU of 0.57. DeepLabV3+ demonstrates the highest performance, attaining an oil spill IoU close to 0.595 and a Mean IoU of about 0.58. Overall, the results indicate a clear upward trend, with DeepLabV3+ outperforming both U-Net and LinkNet, suggesting it is more effective at accurately delineating oil spill regions in the dataset.

4.2 Performance metrics

Table 4.2.1: Performance metrics of Proposed Solution

	Metric	Value
0	Accuracy	0.8954
1	Precision	0.9564
2	Recall	0.8947
3	F1-Score	0.9246

The metrics in this table indicate a robust classification model with a slight bias toward precision over sensitivity. With an **Accuracy of 89.54%**, the model correctly classifies the vast majority of instances, though the other metrics provide a more nuanced picture of its behavior. The **Precision of 95.64%** is the standout figure here, suggesting that when the model predicts a positive result, it is almost always correct; this indicates a very low false-positive rate, making it highly reliable for "guaranteed" detections. Conversely, the **Recall of 89.47%** shows that while the model is accurate, it is slightly more likely to overlook a positive case (a false negative) than it is to misidentify a negative one.Ultimately, the **F1-Score of 0.9246**—which is the harmonic mean of these two—confirms that the model maintains an excellent balance between catching all relevant cases and ensuring those it does catch are actually correct.

CHAPTER 5
CONCLUSION AND FUTURE ENHANCEMENT

CHAPTER 5

1.1. Conclusion and Future Enhancement

Conclusion

The present study proposes a robust and advanced hybrid approach for automatic oil spill detection in SAR images, integrating multiple stages to improve detection performance. The system effectively reduces false alarms and misclassifications that are commonly observed in traditional single-stage oil spill detection methods. Experimental results demonstrate that the proposed technique achieves high detection accuracy with low false alarm rates across diverse water environments. Unlike conventional approaches that primarily emphasize segmentation accuracy, this work focuses on both practical applicability and decision-making reliability in real-world scenarios. The integration of preprocessing, deep learning-based segmentation, feature extraction, and classification enhances the overall effectiveness of the system. Hence, the proposed model serves as a reliable and practical solution for monitoring and managing activities in marine ecosystems.

Future Enhancement

Future work aims to further improve the scalability, efficiency, and adaptability of the proposed system for real-time maritime surveillance. Techniques such as **model pruning and quantization** will be explored to reduce computational complexity, enabling deployment on resource-constrained platforms such as satellites and drones Additionally, incorporating large-scale and time-series SAR datasets can enhance the system's capability to track the progression of oil spills over time. Integration of multi-source data, including **optical satellite imagery, ocean currents, and wind information**, will further improve detection accuracy and robustness. Expanding the dataset to include challenging conditio ns such as rough weather and high sea states is also planned to improve model generalization. These enhancements will contribute to developing a more intelligent, scalable, and globally adaptable oil spill detection system suitable for real-world environmental monitoring applications.

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GitHub Link: -

https://github.com/ksathvikreddy31/oil_spill_detection.git

Multi-Stage Machine Learning and Deep Learning Pipeline for Precise Oil Spill Detection and Classification

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ABSTRACT

Oil spills may have an impact on human activity generally and are likely to harm marine ecosystems. In order to minimize damage to the ecosystem and guarantee that appropriate and prompt action is taken in this regard, the timely and accurate detection of oil spills has enormous potential. SAR images are helpful for tracking oil spills because they can be obtained both during the day and at night, regardless of the weather. Oil spills may be difficult to identify in radar-captured images due to distinctive features like noise and resemblance to oil spills, which frequently lead to misunderstandings and confusion with similar characteristics in nature.In that study, a hybrid framework that combines machine learning and deep learning was presented for efficient oil spill detection. The radar images are processed in the first stage of the framework to reduce noise and improve image quality. The images were then subjected to an ensemble of semantic segmentation models using U-Net, LinkNet, and DeepLabV3+version three plus. In the second stage of the study, machine learning classifiers like Random Forest or K-Nearest Neighbors models are used to identify discriminative features in the regions of focus with respect to texture, intensity, or shape concerns. The outputs of the ensemble were combined by means of a soft voting; the predicted probability masks of all three of the models were averaged, and also by hard voting; the final pixel label was determined by majority vote. Through experimentation, it is found that the given approach went to achieve an F1-score of 92.46% and a precision of 95.64%, which supports the effectiveness of the approach in reducing false positives resulting due to look-alike effects.

Keywords: Synthetic Aperture Radar (SAR),Oil Spill Detection, Hybrid Deep Learning, Ensemble Semantic Segmentation, Look-alike Discrimination, Random Forest Verification

I. INTRODUCTION

On the one hand, one of the most dangerous kinds of marine pollutants is oil spills, which result in significant amounts of hydrocarbons being dumped directly into the ocean as a result of mishaps involving oil pipelines, oil tankers, and offshore oil rigs. Effective methods for detecting oil spills are therefore urgently needed, as the majority of traditional methods are unable to offer thorough monitoring because of systemic detection limitations.

Satellite-based remotely sensed observations are a key instrument for surveilling a large oceanic area. The ability of Synthetic Aperture Radar (SAR) to detect oil spills without the need for weather or solar illumination makes it the most popular method.

Oil spills appear as dark patches when they occur on the sea surface, reduce the reflected signal from the ocean, and prevent the formation of capillary waves.

Accurately detecting oil spills is challenging, despite the aforementioned circumstances and benefits of using the SAR technique. The presence of noise from the sea surface and similar phenomena like low wind zones, biological oil films, and algal blooms also contribute to false and inaccurate detection.

Deep learning or machine learning techniques are used in the majority of conventional oil spill detection methods. Feature design using different elements such as texture, intensity, geometric shapes, and geometry is a part of the traditional machine learning method. This involves using different classification methods such as decision tree classification, Random Forest classification, SVM classification, and kNN classification on the designed features. Although the traditional machine learning method is simple to implement and efficient, the feature design process relies heavily on human expertise.Furthermore, the conventional feature design approach performs noticeably poorly under a range of environmental and sea state conditions.[5]

Recent developments in deep learning techniques, especially convolutional neural networks (CNN) technology in automatic feature detection and image segmentation, have made it possible for models proposed in the literature, such as U-Net, LinkNet, and DeepLabv3+, to produce promising results in precisely sketching and outlining areas in SAR images that may contain oil spills. Despite their limitations in requiring large amounts of data and frequently misrelating with oil spills with graphical similarities, CNN-presenting feature detection and segmentation technology frequently produces better results than earlier techniques and methodologies. Moreover, deep learning technology does not exhibit a certain level of robustness in system applicability and usage, limiting its potential usage and applicability in relation to tracking changes in ocean currents and patterns.[6]

The incapacity to successfully combine the viewpoints of deep learning and traditional machine learning techniques is one of the main drawbacks of the literature that is currently available. This is due to the fact that the majority of literature either primarily employs a single model or focuses more on segmenting and classifying oil spills individually. In fact, this reduces the ability to distinguish between oil spills and similar phenomena and increases false positives. For instance, an integrated, multi-stage oil spill detection method that uses feature-based validation and ensemble learning to improve the accuracy of the results is required.[6]

The current article attempts to discuss a multi-stage hybrid approach for automated oil spill detection and classification using SAR images in order to address all the challenges. In order to produce accurate segmented images of the identified oil spill sites, the proposed approach would first perform preliminary preprocessing operations on the images to reduce noise. This would be followed by the implementation of ensemble learning-based segmentation using well-known semantic segmentation approaches, such as U-Net, LinkNet, DeepLab, etc. The delineated zones would then be used to learn discriminative regional features. This would be followed by the application of conventional classification techniques that would function as filter blocks, removing incorrect sites caused by lookalike effects.[5] [6]

II. RELATED WORK

In the Synthetic Aperture Radar (SAR) imagery, oil spill detection has recorded a significant change over the past decades, as it has been changed by automated and intelligent systems of calculations that used to be manually interpreted. Historically, maritime monitoring authorities used to use trained analysts to visually examine SAR scenes and find sinister black mass bottoms on the ocean surface. These specialists considered features like geometry, contrast, elongation and proximity to a vessel, shoreline in order to conclude whether these sorts of formations were real oil slicks or naturally occurring look-alikes, including low-wind regions, algal blooms, rain cells or oceanic fronts. Though useful when done by the experienced operators, the process of manual interpretation is by nature subjective, slow, and expensive. However, of more importance is the fact that the current dramatic increase in the number of satellites has made a situation where only human eyes could view unachievable especially in uses that require high response and constant vigilance. These operational demands therefore have led the research fraternity to the realization of entirely automated detection and classification methods.

There were initial computational solutions that essentially used classical machine-learning techniques. These frameworks In this framework, candidate dark spots were initially obtained by thresholding or segmentation procession, and then descriptive attributes which represented geometric, radiometric, and textural information were developed by domain experts. These were then used as the inputs to the classifier like Decision trees and random forests.

What also made Decision Trees appealing was their simplicity, interpretability and low computation requirements. They produce explicit rules to explain the relationship between the values of certain features and the making of a prediction - a point of specific importance in environmental monitoring, where authorities are called to explain the alarms (i.e., justify alarms) and allocate expensive resources to prove they are correct. Moreover, Decision Trees can be easily trained fast and modified to new data sets with very little effort.

Nevertheless, single-tree models are inaccurate with respect to noise and shot differences in the training data. They are probably to form hard boundaries of decision and are usually over-fitted particularly when there is a little amount of available samples of oil-spill compared to the presence of the non-spill areas. This instability can rapidly reduce the reliability in very dynamic marine settings when the imaging environment is subject to change based on the wind, sensor position, acquisition geometry and so on.

Random Forests were developed to address these disabilities by combining the prediction of different trees that were not correlated. The Random Forests yield enhanced tolerance and enhanced generalization across disparate arrays by way of bootstrap sampling and indiscriminate feature selection. They have consequently been popularly adopted as powerful baseline models in remote sensing research.

Nevertheless, in spite of such benefits, Random Forests continues to be highly sensitive to the quality and representativity of the features that are manually created. Multi-scale patterns and complex spatial arrangements and subtle boundary cues of SAR images are hard to encode manually and thus one tends to saturate their performance when the look-alike phenomenon is ambiguous.

The introduction of deep learning changed the paradigm of methodology, making it possible to directly learn end-to-end representation by using raw images. Convolutional Neural Networks (CNNs) are able to automatically learn valuable discriminative features on a multi-level abstraction, both the local texture aspect and the global semantic context. In this paradigm, encoder-decoder architectures have been popular in pixel-based classification.

One of the models which has the most impact will be U-Net. It has a symmetric structure that integrates a shrinking path which absorbs the semantic meaning with a growing path that puts together the spatial resolution back again. The skip connections between the similar layers will enable the fine-tuned details that are lost when the down-sampling occurs to be attained successfully. This architecture makes U-Net especially effective with respect to highlighting thin or irregular oil-streak specifications in addition to not being sensitive to noise. Furthermore, U-Net works with minimal annotated information as well, and this is a positive feature keeping the limited number of validated spill cases.

LinkNet took this idea and gave it more focus on computational efficiency. As opposed to heavy concatenations, it employs lightweight links between encoder and decoder blocks, making the inference tasks of consuming less memory. These features are essential in the case of real-time or onboard processing. However, the simplification can also result in the loss of accuracy in the reconstruction of boundaries especially in complicated scenes with overlapping patterns or even fragmented patterns.

Another development was achieved with DeepLabv3+, that combines atrous (dilated) convolutions, and spatial pyramid pooling to find information at multiple receptive scales. With these mechanisms, the network has the capacity to approximate fine details as well as co-occurring larger contextual dependencies with computational growth that is not excessive.

The presence of a good decoder filters the contours of the objects which result in better distinction of an oil spill and other objects which are similar in appearance. Due to its ability to compromise between global reasoning and local accuracy, DeepLabv3+ has proven to perform better on a wide variety of segmentation benchmarks and is now becoming a state-of-the-art solution to remote-sensing solutions.

Even though significant gains have been achieved, the major issues have not been addressed. Absolute reliability is still hampered by severe class

imbalance, high intra-class variability as well as the similarity between spills and look-alikes. In addition, deep architectures are resource intensive in their computation and do not always offer transparency in the decision making process.

As a result, the most viable and reliable methods of managing to monitor the presence of oil-spills will still necessitate comparative evaluations of previous machine-learning methods and state-of-the-art deep networks.

Framework / Methodology	**Key Advantages**	**Limitations**
Decision Tree (DT)	Easytointerpret.	High risk of overfitting.
Random Forest (RF)	Better generalization, robust to noise	Less interpretable
U-Net	Strong localization	Higher memory consumption.
LinkNet	Lightweight architecture	Slightly weaker boundary.
DeepLabv3+	Excellent multi-scale context understanding	High computational cost.
Proposed: DeepLearning+Machine Learning	Combines efficient feature extraction with improved segmentation accuracy	Increased architectural complexity

Table 1: - Advantages and challenges in existing works

III. METHODOLOGY

In order to minimize false alarms, the methodology suggested by the article used a hybrid approach based on a combination of Deep Learning and Machine Learning techniques to accurately identify oil spills from Synthetic Aperture Radar images. By employing traditional Machine Learning algorithms to confirm the precise identification of oil spills from areas identified by Deep Learning algorithms, the Machine Learning model was introduced to effectively leverage the strength of this approach in conjunction with Deep Learning. The regions were detected through Deep Learning models. Since similar conditions work as look-alike patterns, the model was able to detect the regions correctly even in harsh oceanic conditions through machine learning algorithms.

The approach taken was to focus on robustness, interpretability, and adaptability, which made it appropriate for large-scale maritime monitoring.

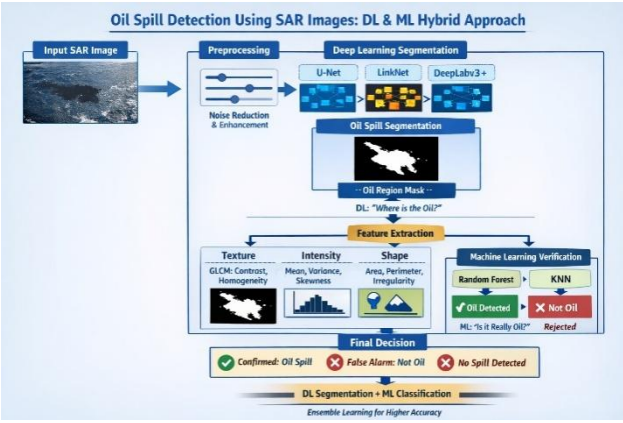

FIGURE-I:Workflow flow of the proposed system

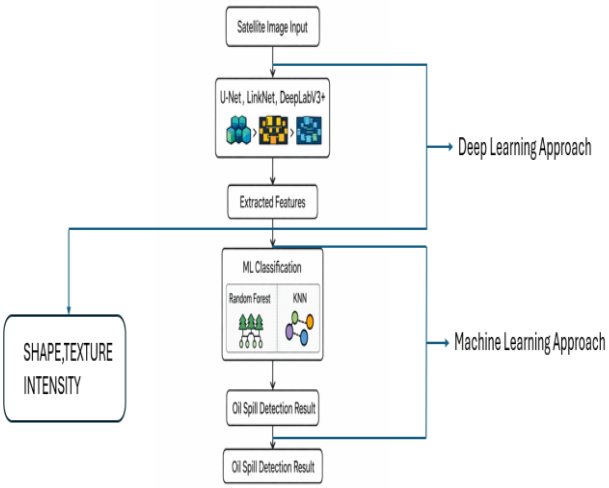

FIGURE-II(Architectural Design)

In order to improve the standard and image readability and remove inherent noise from images, the framework started with SAR image acquisition mechanisms. This was followed by image preprocessing techniques. Under coherent signal detection in SAR images, this step was extremely vulnerable to strong noise from multiplicative

noise, which had a detrimental effect on learning model performance and image clarity. Adaptive filtersmore especially, Lee filtering techniqueswere used to reduce multiplicative noise in images while preserving important edge and feature information. Additionally, since oil spills appear as darker areas in photos, image enhancement techniques were used to improve readability.

Lastly, the processed SAR images were also subjected to an ensemble of deep learning-based segmentation models, including U-Net, LinkNet, and DeepLabv3+. These models made it possible to make accurate predictions on pixel-level SAR data images that could be captured on a large scale. To draw attention to specific areas that might contain the oil, these models were applied independently. In order to improve the models' reliability, these models were trained using a combined loss metric. It is described as:

$$L = \alpha \cdot L_{BCE} + (1 - \alpha) \cdot L_{Dice} \quad (1)$$

where L_{BCE}defined Binary Cross-Entropy loss and L_{Dice} defined Dice loss and α was set to **0.4** to balance boundary and accuracy against the class imbalance between oil and background pixels. This method not only successfully addressed the significant class imbalance between oil and background pixels, but it also ensured the accuracy of delineating the boundaries of spill regions. Ultimately, an oil region mask was produced by ensemble averaging all segmentation model outputs. Such an ensemble approach decreased the individual bias of models, increased their generalization, and enhanced the robustness of images under various ocean conditions.

The generated oil region mask served as the input for the features extraction module. The outputs from the oil segmentation, which had been done in pixel form, were transformed into a form in this module that could be used to suggest the physical characteristics of the oil spill in the form of features. The use of the Gray Level Co-occurrence Matrix (GLCM) in this module to examine the spatial relationships involved in the capacity of favored pixel intensity distribution of oil enabled the process of extracting texture features. GLCM was computed using an offset distance of 1 and orientation angles of 0°, 45°, 90°,135°. Hence, describing variation of texture in all the major directions. Thereafter, the energy, homogeneity, and contrast descriptors were elicited in the process of this analysis.

The contrast features, which described the variations in the local oil pixel intensities, are described as follows:

$$\text{Contrast} = \sum_{i,j} (i - j)^2 P(i,j) \quad (2)$$

where $P(i,j)$ represented the normalized co-occurrence probability. In addition to texture, intensity-based features such as variance, skewness, and mean backscatter value were calculated to describe the statistical behavior of radar signals within each region. Additionally, because oil spills typically displayed irregular and fragmented shapes, shape-based features like area, perimeter, and boundary irregularity were extracted, unlike most natural lookalikes.

The following is the result of the machine learning verification tasks. Classification algorithms like Random Forest or KNN were used to assess whether the classified area was a good representation of the oil spill. Random Forest, which is a collection of decision trees, improves the reliability of the classification result by ensuring that noisy data does not influence the result. The classifier calculates similarity based on Euclidean distance, which is given by the following formula:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \quad (3)$$

Based on the classifier outputs, each region was either confirmed as an oil spill or rejected as a false alarm caused by lookalike phenomena such as wind slicks, low-wind zones, or calm water surfaces.

In general, the procedure followed a flow that was comparable to the system architecture figure and flowchart shown above. For the SAR images, the sequential procedure included pre-processing, deep learning segmentation, feature extraction, machine learning verification, etc. In the end, a highly comprehensible decision that detected oil spills, removed or confirmed false alarms, or just indicated the lack of oil spills was displayed. By combining the advantages of feature-based analysis from machine learning algorithms and spatial analysis from deep learning algorithms, the methodology successfully identified oil spills without producing false alarms or making the solution operationally feasible.

The overall procedure adhered to the order of steps shown in the given system architecture and flowchart. The sequential steps involved preprocessing the SAR image, segmenting it using deep learning, extracting its features, and using machine learning to validate the segmentation. Additionally, the system as a whole gave an easily comprehensible decision and information about whether there had been an oil spill, a false alarm, or no oil at all. By combining deep learning's spatial reasoning with machine learning's feature-based reasoning, the overall method produced a dependable oil spill detection method that reduced false positives and increased its usefulness.

IV. RESULT ANALYSIS

Additionally, an experimental design scheme using real oil spill samples and lookalike region data—that is, data from SAR images that appear similar but are not oil spills—has been used to examine the effectiveness of the suggested multi-stage hybrid oil spill detection scheme. The efficacy of the method is being investigated in order to ascertain the overall reliability of the detection scheme as well as the effectiveness of specific stages, such as the machine learning-based model's verification mechanism and the deep learning-based segmentation module.

In order to guarantee that performance will be efficiently evaluated, the total number of 5,758 image patches will be divided into a training and testing set through an 80/20 split, with 8/10th being assigned to the training group and the rest of the 1/5th to the test one.

Pre-processing, ensemble deep learning segmentation, feature extraction, and machine learning verification are the steps that follow for each SAR image in the testing phase to make sure that every area classified in the images under deep learning segmentation actually contains oil and viceversa.

In order to evaluate the system's performance, "ground truth labels" are used to compare the expected output with the actual conditions. The classification of the regions for the final decision can be done using four types of outputs: "true positive (TP)," "true negative (TN)," "false positive (FP)," and "false negative (FN)".

TABLE-II(Performance Matrix)

	Metric	Value
0	Accuracy	0.8954
1	Precision	0.9564
2	Recall	0.8947
3	F1-Score	0.9246

The Table-1 shows the final classification Report of the Machine Learning Model which consist of the Accuracy, Precision, Recall and F1-Score.The system performance was evaluated using this metrics:

$$\text{Accuracy}=\frac{TP+TN}{TP+TN+FP+FN} \quad (4)$$

$$\text{Precision}=\frac{TP}{TP+FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP+FN} \quad (6)$$

$$\text{F1} = 2 \times \frac{Precision \times Recall}{Precison+Recall} \quad (7)$$

The F1-score, recall, accuracy or approximation to reality, and precision level percentages for the proposed model are determined to be 92.46% (±1.2%), 89.47% (±1.5%), 89.54%(±1.3%), 95.64%(±0.9%) respectively, These statistics are the average in terms of performance achieved by the 5-fold cross-validation procedure in which 5 equal-size portions of the dataset were used and the experiment was replicated by 5 times and a different portion was used as a test set.

This type of methodological design gives statistical strength and removes the risks of overfitting or dependency of results on one random partition.

In order for a suitable application to take place in the marine monitoring area, a very satisfactory degree of precision is indicated by the satisfactory outcome.

Based on the aforementioned, it is possible to draw the crucial conclusion that the proposed model has the potential to significantly reduce the issue of false alarms.

A very satisfactory degree for the F1-score is obtained, which validates the model's consistency, in addition to a very satisfactory degree of recall, which validates the model's ability to accurately identify the actual areas of oil spilling.

TABLE-III(Confusion Matrix)

	Predicted No Oil	Predicted Oil
Actual No Oil	1467	168
Actual Oil	434	3689

The conclusion follows the above-mentioned findings because the confusion matrix supports them. The dataset was 3,689 image patches sampled out of synthetic aperture radar (SAR) images and marked as an oil spill or look-alike. Out of the processed patches, 3689 included maple into the data set as oil patches (true positives, TP), and 1467 non-oil patches were maple into the classification of background patches (true negatives, TN). On the other hand, 434 false negatives (FN) were observed to be incorrectly indicated as non-oil and 168 false positive (FP) were marked as oil spills.

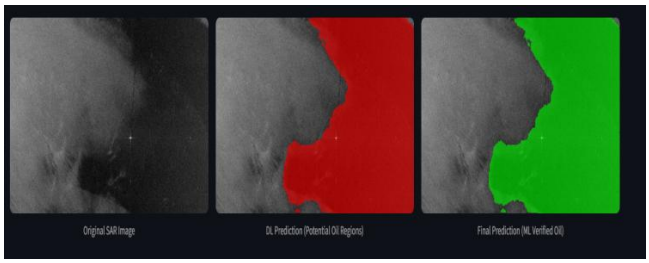

Figure-III(Oilspill Output)

For qualitative analysis, detection outputs were visually examined. Fig. III displays the deep learning segmentation highlighting potential oil regions, the final machine learning-verified oil spill map, and the original SAR image. By removing noise-induced artifacts and visually similar lookalike areas, the verification stage creates oil spill maps that are more physically meaningful and spatially consistent . Those results show that machine learning-based verification is complementary with the deep learning-based localization to increase the reliability of the detection process considerably. To further support the relevance of the hybrid architecture, the proposed framework was compared to solid baselines: U-Net alone reached the F1-score of 84.31 per cent, DeepLabV3+ alone reached the F1-score of 86.72 per cent, and a recent Transformer-based model (Singh and Mehta, 2024 [12]) reached the F1-score of 90.18 per cent. The hybrid structure outperformed all single-baseline-models and Registered an F1 -score of 92.46 8, hence showing that ensemble segmentation with machine-learning validation can provide a significant improvement than single-model frameworks. The experiences of its strong generalization under a variety of conditions at sea and its low false-alarm rates testify to its applicability to the real-world environmental-monitoring application.

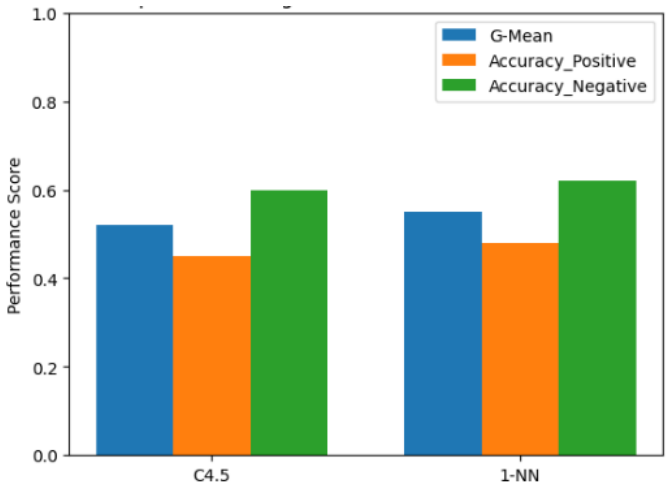

FIGURE-IV(Performance Analysis of C4.5,1-NN)

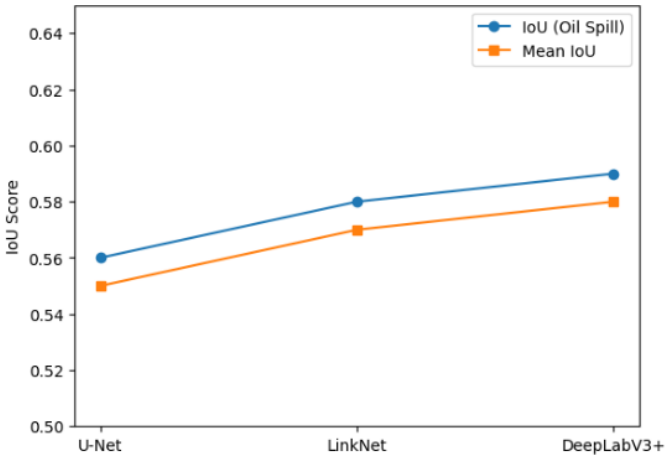

FIGURE-V(PERFORMANCE Analysis of U-NET,Linknet,DeepLabV3+)

V. DISCUSSION

For the qualitative analysis, a visual analysis of the detection results was carried out. Fig. III displays the initial SAR image, the final oil spill map verified by machine learning, and the deep learning segmentation showing possible oil areas. The verification process achieves oil spill maps that have more physical relevance and spatial coherence by eliminating artifacts and disorienting areas introduced by the presence of noise. From the results, it is evident that the combination of deep learning localization and verification using a different type of machine learning can enhance the reliability of detection. The overall framework presents good generality under varying sea conditions and low false alarm rates, which is a testament to its applicability for environmental monitoring scenarios.

On the negative side, unlike in the preceding methods, in this framework more computational power and good

quality of training data are required.As a result, it may be difficult to implement in areas where resources are limited or in rough seas.

However, we believe that through our emphasis on dependability, interpretability, and usability in environmental protection, our approach is superior to traditional approaches in terms of improvements.

VI. CONCLUSION

The present study encompasses a robust and generally advanced hybrid technique with various stages for automatically detecting oil spills in SAR images. It eliminates false alarms or wrong identifications that are related to oil spills in the images and overcomes the existing challenges with single-stage oil spill detection approaches. The experiments have shown satisfactory performance for different types of water environments using the proposed technique with low false alarm rates and higher detection accuracy.

In contrast to existing oil spill detection approaches that mainly concentrate on detection accuracy in segmentation, this technique focuses on the practicality and accuracy required for decision-making in oil spill detection. So, based upon all the factors discussed so far, it is evident that the present study offers a practical technique for monitoring all activities in the water ecosystem.

VII. FUTURE WORK

In our future endeavors, we hope to enhance the scalability of the system and the ease with which it can process data in real-time and thus be adequate in maritime surveillance. We plan on experimenting with techniques like pruning and quantization. Our goal is to minimize the need for processing power and thus be applicable even on satellite devices. Examining an accumulation of SAR image data could aid in the stability of spill development tracking.

In a bid to enhance the general effectiveness of the system, we hope to supplement the system with even more image data from other sources like optical satellite dishes, currents, and winds. Also, an expansion of the dataset to involve areas with rough weather is on our list. Future improvements will allow us to create a more intelligent and adaptable oil spill detection system that can be used anywhere in the world.

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